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Exact degree-ten invariants of a self-dual five-form in ten dimensions

[Author list pending mentor review]

Abstract: A self-dual five-form in ten dimensions carries a 126-dimensional space of field values, and the Lorentz-invariant scalar functions built from it organise the possible non-linear self-interactions of the field. We determine the degree-ten sector of this invariant structure exactly.

Working in two independent formulations — a tensor formulation built from contraction graphs, and a spinor formulation built from chiral bispinors — we construct an explicit map between them and certify that it is injective with image the gamma-traceless subspace, of rank 126, with an exact left inverse — a left inverse, not a two-sided one, since the map is not onto the full symmetric square. The two formulations are then compared degree by degree on common samples, and their spans agree at degrees four, six, eight and ten.

Three results are established exactly over the rationals rather than modulo a prime. The full degree-ten invariant space has dimension 14. The subspace reachable by the stress-tensor flow has dimension 11, computed by running the closure to its fixed point in exact rational arithmetic after lifting its non-integral targets by rational reconstruction. The quotient of the first by the second therefore has dimension 3: there are three independent degree-ten invariant directions that this flow does not reach. We also show that the span of the previously published degree-ten structures meets the product sector in exactly one dimension, and we write that intersection out as an explicit integer identity.

For the generic functional rank of the full candidate family we certify the lower bound 81 by an exact modular Jacobian computation with an explicit 81×81 minor; the matching upper bound is analytic and is due to earlier work, which we cite rather than reprove.

All results are supported by machine-checked certificates, and the computations are reproducible from a public repository. We state explicitly which claims are theorems, which are exact certificates, which rest on agreement across several primes, and which remain open.

Contents

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1 Introduction

A chiral, or self-dual, p-form field is one whose field strength is constrained to equal its own Hodge dual. The constraint is natural and it is forced on us in several places in string theory, but it sits awkwardly with the usual machinery of field theory: a self-dual field strength cannot be derived from a manifestly Lorentz-invariant action in the naive way, a difficulty identified by Marcus and Schwarz [1] and since addressed by a sequence of increasingly satisfactory constructions — the non-manifestly covariant Hamiltonian formulation of Henneaux and Teitelboim [2], the duality-symmetric actions of Schwarz and Sen [3], the auxiliary-scalar formulation of Pasti, Sorokin and Tonin [4, 5], the decoupled-auxiliary-field action of Sen [6], and the covariant polynomial actions of Mkrtchyan [7]. The constraints on *interacting* chiral forms were sharpened early by Perry and Schwarz [8], whose six-dimensional Born–Infeld analysis supplies the condition a self-interacting chiral form must satisfy, and by Bekaert and Henneaux [9], who derive that condition from the Dirac–Schwinger requirement on the energy–momentum tensor and set out the topological subtleties in splitting a non-chiral form into two chiral halves.

Once one asks not merely for *a* self-dual theory but for the space of *non-linear* self-interactions a chiral form can have, the question becomes one about scalar invariants. A Lagrangian, or in formulations where no Lagrangian is available a deformation of the duality constraint, is built from Lorentz-invariant scalar functions of the field strength. Knowing which such functions exist, how many are independent, and how they are related is therefore a prerequisite to classifying the theories. In four dimensions this is easy: there are two invariants, the story is classical, and even the non-linear duality-invariant theories that extend Maxwell's equations while preserving conformal invariance have been completely characterised [10], and subsequently extended to a supersymmetric setting [11]. In six dimensions the analysis of self-interacting chiral two-forms is by now well developed [12, 13], and the interplay with $T\bar{T}$-like stress-tensor flows is understood in some detail [14].

The stress-tensor flow that supplies the deformation studied here has its own lineage. The $T\bar{T}$ operator is well defined as a composite because its expectation value factorises [15]; the flow it generates was identified as a solvable direction in the space of two-dimensional theories by Smirnov and Zamolodchikov [16] and, independently, by Cavaglià, Negro, Szécsényi and Tateo [17], and it admits a geometric reading as a field-dependent change of coordinates [18]. What we use at degree ten is the higher-dimensional analogue of the flow, not these two-dimensional results themselves; they are cited to fix what the construction is modelled on.

Ten dimensions is harder, and the difficulty is not conceptual but combinatorial. The self-dual five-form lives in a 126-dimensional module of $\mathrm{Spin}(1,9)$, the invariants of low degree are already numerous, and the number of contraction patterns one can write down at degree ten is large enough that hand enumeration is unreliable. Cederwall, Hutomo, Kuzenko, Lechner and Sorokin [19] give explicit degree-ten structures and analyse the invariant content of the five-form; the companion paper of Hutomo, Lechner and Sorokin [20] develops the non-linear chiral four-form theories in $D = 10$ and the stress-tensor flow whose reach is one of the objects we study here. What those works leave open, and what this paper supplies, is the exact dimension of the degree-ten invariant space, the exact dimension of the part of it the flow reaches, and the exact relation between the published structures and the products of lower invariants.

Prior computational method. The general workflow we use — enumerate contraction graphs, evaluate the resulting invariants on generated tensor data, and read off the linear relations from the ranks — is not new, and we do not claim it. It is the method of

Elamaran, Ferko and Scarlett [21], who introduce it and demonstrate it on a three-form in six dimensions. Our contribution on the computational side is not the workflow but the certification: we replace floating-point rank determination with exact arithmetic, and we attach to each dimension a certificate that a reader can check independently.

Why exactness matters here. A rank computed in floating point is a judgement about how small a singular value has to be before it counts as zero. For the questions in this paper that judgement is exactly the thing at issue: the difference between $\dim Q_{10} = 3$ and $\dim Q_{10} = 0$ is the difference between three invariant directions the flow misses and none, and an earlier stage of this project got that answer wrong for reasons that a tolerance would have concealed rather than revealed (section 9). Every rank quoted below is therefore computed exactly, either in a finite field or over $\mathbb{Q}$.

Contributions.

1. An explicit tensor–spinor bridge for the ten-dimensional self-dual five-form, certified to have rank 126 onto the gamma-traceless symmetric chiral bispinors, with an exact left inverse, a verified round trip and verified equivariance (section 5).
2. A degree-resolved verification that the tensor and spinor formulations span the same invariants at degrees four, six, eight and ten (section 7).
3. The exact rational dimensions $\dim_{\mathbb{Q}} A_{10} = 14$, $\dim_{\mathbb{Q}} D_{10} = 11$ and $\dim_{\mathbb{Q}} Q_{10} = 3$, with explicit minors and annihilating covectors (section 9).
4. An analytic derivation that the free stress-tensor trace vanishes at quadratic order, together with a counterfactual showing the derivation is load-bearing: assuming otherwise returns $Q_{10} = 0$ (section 10).
5. The exact intersection $\dim_{\mathbb{Q}}(B_{10} \cap P_{10}) = 1$, written out as an integer identity (section 11).
6. An exact certification of the generic functional rank 81 as a lower bound, over the complete 83-candidate family, with an explicit 81×81 minor (section 8).
7. A reproducibility architecture in which every number in this paper is generated from a certificate rather than transcribed (section 12).

2 Ten-dimensional self-dual five-forms

We work on $\mathbb{R}^{1,9}$ with metric $\eta = \mathrm{diag}(-1, +1, \ldots, +1)$ and a fixed orientation, with volume form $\epsilon_{\mu_1\ldots\mu_{10}}$ normalised by $\epsilon_{01\ldots9} = +1$. The Hodge star acts on a five-form by

$$(\star F)_{\mu_1\ldots\mu_5} = \frac{1}{5!}\,\epsilon_{\mu_1\ldots\mu_5}{}^{\nu_1\ldots\nu_5} F_{\nu_1\ldots\nu_5}. \tag{2.1}$$

In Lorentzian signature and for a middle-degree form in ten dimensions $\star^2 = +1$, so the star has real eigenvalues ±1 and the space of five-forms splits,

$$\Lambda^5 = \Lambda^5_+ \oplus \Lambda^5_-, \qquad \dim \Lambda^5 = 252, \qquad \dim \Lambda^5_\pm = 126. \tag{2.2}$$

A self-dual five-form is an element of Λ^5_+, cut out by the projector

$$P_\pm \;=\; \tfrac{1}{2}\big(\mathbf{1} \pm \star\big), \qquad P_\pm^2 = P_\pm, \qquad P_+ + P_- = \mathbf{1}, \qquad \operatorname{tr} P_\pm = 126. \tag{2.3}$$

Throughout, F denotes an element of Λ^5_+ and the whole discussion is about polynomial functions of its 126 components.

That $\star^2 = +1$ rather than -1 is what makes (2.3) a real projector, and it is a coincidence of signature and dimension rather than a general fact. For a p-form in d dimensions with $d = 2p$ and metric signature (t, s),

$$\star^2 \;=\; (-1)^{p(d-p)+t} \;=\; (-1)^{p^2+t} \qquad \text{on middle-degree forms,} \tag{2.4}$$

which for $p = 5$, $d = 10$ gives $\star^2 = -(-1)^t$: real eigenvalues ± 1 in Lorentzian signature ($t = 1$), and eigenvalues $\pm i$ with no real eigenspace decomposition in Euclidean signature ($t = 0$). Self-duality in ten Lorentzian dimensions is thus available for a five-form and unavailable for its Euclidean counterpart, and (2.4) is asserted by test wherever the decomposition is used rather than assumed — this project once mistook a Euclidean projector for an anti-self-dual one and briefly read the resulting non-vanishing as signal.

This is also the source of the difficulty that motivates the whole subject. The self-duality constraint $\star F = F$ is a first-order condition on the field strength, and imposing it on a p-form with p odd obstructs the naive construction of a manifestly Lorentz-invariant action, as Marcus and Schwarz [1] observed: the kinetic term $\langle F, F\rangle$ vanishes identically on the constrained surface, by exactly the computation carried out in (10.3) below. Every construction cited in section 1 is a way around that vanishing.

The Lorentz group acts on Λ^5_+, and the objects of interest are the polynomial functions invariant under that action. We grade them by degree in F; because F appears polynomially and the invariants are built by full contraction, only even degrees occur, and we write A_d for the space of invariants of field degree d. This paper is about A_{10}, with A_4, A_6 and A_8 appearing as inputs and as consistency checks.

Generic orbit dimension. The Lorentz algebra has dimension 45. A generic point of Λ^5_+ has a stabiliser, and the dimension of the generic orbit is 45 minus the dimension of that stabiliser. The count of functionally independent invariants at a generic point is correspondingly 126 minus the generic orbit dimension. This analytic count is 81 and it is not ours; it appears in the literature we build on [20]. What we contribute is an exact computational lower bound matching it, discussed in section 8.

Real and complex statements. It matters, and it will matter again in section 5, whether a given statement is about a real form or about the complexification. Dimension counts of invariant spaces are statements about the complexified problem and are insensitive to the real form. Statements about which eigenspace of $\star$ is which, by contrast, are not: they depend on signature and orientation, and getting them wrong is a silent error rather than a loud one.

3 Tensor formulation

In the tensor formulation an invariant is specified by a contraction pattern. Write F with its five indices, take some number of copies, and contract all indices in pairs using η. Each such pattern is naturally encoded as a graph: vertices are copies of F, and edges are index contractions, with each vertex carrying five ports.

Two different-looking contraction patterns can define the same invariant, so the graphs must be reduced to a canonical form before they are counted. We do this with the `nauty` canonical labelling of McKay and Piperno [22], reached through the `pynauty` bindings. This step is not a convenience: an earlier version of this project used an inexact hashing fallback which collided, mapping distinct classes onto the same key and undercounting the atlas. Canonicalisation here is exact at every degree used.

Two structured tensors recur and are worth naming. From two copies of F one builds the symmetric tensor

$$M_{\mu\nu} = F_{\mu\alpha\beta\gamma\delta}\, F_{\nu}{}^{\alpha\beta\gamma\delta}, \tag{3.1}$$

whose traces $\mathrm{Tr}(M^k)$ furnish a convenient family of invariants, and from four copies one builds the two irreducible structures we label $N^{(1050)}$ and $N^{(4125)}$ after the dimensions of the Spin(10) representations in which they transform. Index placement conventions for all three are fixed in appendix D; they are stated there rather than here because nothing in the argument depends on the choice, only on its consistency.

We distinguish throughout between *product* invariants — those that factorise as a product of invariants of strictly lower degree, such as $I_1^{(4)} I_1^{(6)}$ at degree ten — and *primitive* invariants, which do not. The product subspace at degree ten is written P_{10}; the span of the graph generators is written G_{10}. Table 1 collects the notation.

symbol	meaning	value
d	spacetime dimension	10
F	five-form field strength	$F \in \Lambda^5$
$\star$	Hodge star on middle forms	$\star^2 = +1$
$\Lambda^5_\pm$	(anti-)self-dual eigenspaces	dim $= 126$
A_d	full invariant space at field degree d	—
G_{10}	graph-generator subspace	—
P_{10}	product subspace	—
B_{10}	span of the published structures	—
D_{10}	subspace reachable by the stress flow	—
Q_{10}	quotient A_{10}/D_{10}	—
τ	stress tensor of the deformed theory	—
S_+	chiral spinor module	dim $= 16$

Table 1. Notation and conventions used throughout. Entries marked "—" are defined in the text rather than by a single number.

4 Spinor formulation

The second formulation replaces the five-form by a bispinor. In ten dimensions the chiral spinor module S_+ has dimension 16. The symmetric square $\mathrm{Sym}^2 S_+$ has dimension 136, and it carries a gamma-trace map whose image accounts for 10 of those dimensions. The kernel of the gamma trace therefore has dimension $136-10=126$, matching $\dim \Lambda^5_+$ exactly. This coincidence of dimensions is the reason a bridge between the two formulations can exist at all; the Clifford-algebra facts behind it, including the existence of Majorana–Weyl spinors in this signature, are standard and we follow the conventions of Kugo and Townsend [23].

Invariants in this formulation are built by contracting copies of the bispinor with gamma matrices and charge-conjugation matrices along closed index paths. We organise these as *port graphs*, the spinor-side analogue of the tensor contraction graphs. In addition to the port graphs we admit a family of *tensor words*: expressions in which a tensor-side structure is used as an intermediate. Whether the tensor words are needed is not an aesthetic question, and section 7 shows that at degree eight they are: removing them lowers the rank. The candidates denoted Ω and Θ in the detailed inventory of appendix E belong to this family.

Full conventions — chirality, charge conjugation, the symmetry properties of $C\Gamma^{(k)}$ and the normalisation of the five-gamma — are in appendix A.

5 Real forms and the exact tensor–spinor bridge

5.1 The map

The bridge sends a self-dual five-form to a symmetric chiral bispinor by contracting with the five-index gamma:

$$\Phi : \Lambda^5_+ \longrightarrow \mathrm{Sym}^2 S_+, \qquad \Phi(F)_{ab} = \frac{1}{5!}\, F_{\mu_1\cdots\mu_5}\, (C\Gamma^{\mu_1\cdots\mu_5})_{ab}. \tag{5.1}$$

That $C\Gamma^{(5)}$ is symmetric in ab, so that the image lies in the symmetric square, is a property of the ten-dimensional Clifford algebra; it is verified directly in the implementation rather than assumed.

The target of (5.1) is not all of $\mathrm{Sym}^2 S_+$. Writing the gamma trace as the contraction

$$(\gamma\text{-tr}\,\psi)_\mu \;=\; \eta_{\mu\nu}\,(\Gamma^\nu)^{ab}\,\psi_{ab}, \qquad \psi \in \mathrm{Sym}^2 S_+, \tag{5.2}$$

a dimension count fixes the image exactly. With $\dim S_+ = 16$ we have $\dim \mathrm{Sym}^2 S_+ = \frac{1}{2}16(16+1) = 136$, while (5.2) has rank 10, one for each value of μ. Hence

$$\dim\ker(\gamma\text{-tr}) \;=\; 136-10 \;=\; 126 \;=\; \dim\Lambda^5_+, \tag{5.3}$$

and the two spaces have equal dimension. Equality of dimensions is of course not equality of spaces, and the substance of the following proposition is that the map realises the isomorphism the count permits.

Proposition 1 (Certified properties of the bridge)**.** *Over* $\mathbb{F}_p$*, for each prime used, the map* (5.1) *satisfies*

$$\operatorname{im}\Phi = \ker(\gamma\text{-tr}), \qquad \operatorname{rank}\Phi = 126, \tag{5.4}$$

$$\ker\Phi \cap \Lambda^5_+ = 0, \qquad \Phi\big|_{\Lambda^5_-} \equiv 0, \tag{5.5}$$

$$\Phi^-\Phi = \mathrm{id}_{126}, \qquad \Phi\,\Phi^- = \Pi_{\gamma\text{-tr}}, \tag{5.6}$$

where $\Pi_{\gamma\text{-tr}}$ *is the projector onto* $\ker(\gamma\text{-tr})$*, and, for* Λ *in the Lorentz group with spinor representative* $S(\Lambda)$*,*

$$\Phi(\Lambda\cdot F) = \chi(\Lambda)\, S(\Lambda)\,\Phi(F)\,S(\Lambda)^T, \qquad \chi \equiv 1. \tag{5.7}$$

Three of these deserve comment, since each is a place where a weaker statement would have been easy to mistake for the real one.

The second identity in (5.6) is not a typographical variant of the first. Φ has full column rank but is not surjective onto $\mathrm{Sym}^2 S_+$, so it admits a left inverse and no two-sided one; $\Phi\Phi^-$ is a projector of rank 126, not the identity on the 136-dimensional space. Both halves are checked as matrix identities rather than on sampled vectors, because an identity that holds on a handful of random F and fails on a subspace is exactly the failure sampling cannot see.

The second identity in (5.5) is the sharp one. On the opposite eigenspace the map does not merely lose rank, it vanishes identically. This is forced by the chirality structure: on S_+ the anti-self-dual part of $\Gamma^{(5)}$ annihilates the module and only the self-dual part acts, so Φ is blind to Λ^5_- by construction. That blindness is precisely what makes an orientation error silent, and is the subject of section 5.3.

In (5.7) the character χ is *solved for* rather than assumed. Asserting $\chi = 1$ and checking the resulting identity would test nothing, since any discrepancy could be absorbed into an unverified constant; solving for χ and finding it identically unity is a check that can fail. Each statement in the proposition is verified by an explicit test at every prime used, with no floating-point tolerance anywhere; the inventory is in appendix C. What we claim is the certification, not the formula: (5.1) is standard.

5.2 Which real form, and why it is not a frame choice

The spinor-side computation is naturally carried out in a null, or oscillator, frame, which is adapted to split signature $(5,5)$. The physical problem is Lorentzian, signature $(1,9)$. These two real forms share a complexification, $\mathfrak{so}(10,\mathbb{C})$, and every dimension count in this paper is a statement about that complexification, so no result depends on which real form the arithmetic is done in. Precisely,

$$\mathfrak{so}(1,9)\otimes\mathbb{C} \;\cong\; \mathfrak{so}(10,\mathbb{C}) \;\cong\; \mathfrak{so}(5,5)\otimes\mathbb{C}, \tag{5.8}$$

and every dimension, rank and intersection reported below is the dimension of a $\mathbb{C}$-linear object, hence constant along (5.8).

We state one thing explicitly because an earlier version of this project recorded the opposite and it took some effort to correct: the Lorentzian and split real forms are *not* related by a real orthogonal frame transformation. There is no $\Lambda \in O(1,9;\mathbb{R})$ carrying $\eta_{(1,9)}$ to $\eta_{(5,5)}$, since a real congruence preserves the signature; the two are different real forms of the same complex algebra, and only after complexifying are they identified. What (5.8) transports is dimension counts. What it does not transport is the question of *which* eigenspace of $\star$ survives, because $\star$ is defined with η and the reality of its eigenspaces is exactly a signature-dependent statement — the subject of the next subsection.

5.3 The orientation branch, and a prime that exposed it

Implementing the null frame over $\mathbb{F}_p$ requires solving a congruence for the frame matrix L, and that solution involves a square root. A square root over $\mathbb{F}_p$ has two branches. The two branches differ by an orientation reversal, an orientation reversal flips the sign of the Hodge star, and flipping the sign of the star exchanges Λ^5_+ with Λ^5_-. The consequence is that with the branch left unpinned, the construction computes the self-dual projector at some primes and the *anti*-self-dual projector at others, with nothing to announce the difference.

This is not hypothetical. Three cells of the rank matrix of section 8 failed at $p = 32707$, at every seed. The natural first hypothesis — an exceptional prime, to be excluded — is wrong, and it is worth recording how we know. Flipping the branch by hand at $p = 32707$ restores the expected behaviour, and flipping it at a prime that had been working breaks that prime in exactly the same way. The failure tracks the branch, not the prime.

The resolution was to pin the branch by construction rather than to exclude the prime. The frame routine now selects, deterministically, the branch under which the self-dual space is the surviving one, and raises an error if neither branch works rather than guessing.

The mechanism is worth writing down, because the failure is silent. The null frame is a matrix L over $\mathbb{F}_p$ solving the congruence

$$L^T \eta \, L \;=\; \eta_{\text{null}}, \tag{5.9}$$

and (5.9) determines L only up to the square-root branches taken while solving it. Two solutions L and L' of (5.9) with $\det L' = -\det L$ are both legitimate frames, but they induce $\star \mapsto +\star$ and $\star \mapsto -\star$ respectively, and therefore

$$\Phi\big|_{\Lambda^5_+} \text{ has rank } 126 \quad \text{versus} \quad \Phi\big|_{\Lambda^5_+} \;\equiv\; 0, \tag{5.10}$$

by (5.5). Nothing in the arithmetic distinguishes them: the wrong branch raises no error and returns a perfectly well-formed matrix of the wrong eigenspace.

One caution about what this selection does and does not verify. The criterion used to *choose* the branch is that the composite have rank 126, so reporting that rank afterwards would be circular as a check even though the construction is sound. The independent check is a different one: on the image of the selected construction we verify directly that $\star F = +F$, that is, that the surviving eigenvalue is $+1$ and not -1. That is a property of the output, not the criterion that produced it, and it is what ties the implementation to the orientation convention fixed in appendix A.

Remark 2. No number in this paper changed as a result of this fix. Every certificate had been computed at a prime where the unpinned code happened to select correctly, and recomputation after the fix returned identical values. What changed is that the stated convention is now true by construction rather than true at the primes that happened to be used. We report it because a convention that holds by accident is a defect even when it has not yet produced a wrong number.

6 Exact computational methodology

Two independent evaluators. The tensor and spinor sides are implemented separately, with no shared evaluation code. This is what makes the agreement of section 7 informative: had one evaluator been a wrapper around the other, agreement would have been a tautology.

Exact modular evaluation. Invariants are evaluated at sample points with coordinates in $\mathbb{F}_p$ for primes near 2^{15}, with reduction after every contraction step so intermediate values cannot overflow. Ranks are obtained by exact Gaussian elimination over $\mathbb{F}_p$. Contraction order is chosen by a greedy planner supplied by `opt_einsum` [24], which is a requirement rather than an optional accelerator: it also supplies the intermediate-size and floating-point-operation estimates that the contractor enforces as a hard budget, so without it that budget is not checked at all (appendix L). Array handling throughout is NumPy [25].

What a modular rank does and does not establish. A rank computed mod p is a lower bound on the rank over $\mathbb{Q}$, and equals it unless p divides a relevant minor. Agreement across several primes makes an unlucky coincidence implausible but does not prove anything. Two arguments close this gap in the cases that matter.

First, the spanning-set argument. If a space is spanned by exactly k explicit elements then $\dim_{\mathbb{Q}} \leq k$ for free, so a modular rank of k pins the rational dimension with no prime excluded. Applying this requires the spanning elements to be defined over $\mathbb{Q}$, and it is worth being explicit that they are. An invariant here is specified by a contraction graph — an integer incidence pattern — and evaluating it contracts copies of F with η and ϵ, both of which have integer entries. Each candidate is therefore a polynomial with *integer* coefficients in the 126 coordinates, defined before any prime is chosen; the prime enters only when that fixed polynomial is evaluated. So A_{10}, G_{10} and P_{10} are each spanned by an explicit integer-coefficient family, and the argument applies to them. For B_{10} the published structures are lifted to $\mathbb{Q}$ outright (appendix J), so it is covered twice over.

Second, where no spanning-set argument is available — for D_{10}, and through it Q_{10}, since the closure produces its elements rather than being handed them — we lift to $\mathbb{Q}$ explicitly, by the Chinese remainder theorem and rational reconstruction in the sense of Wang, Guy and Davenport [26], and validate the lift at a prime held out of the fit. Sampling arguments elsewhere in the pipeline are of Schwartz–Zippel type [27, 28].

Analytic derivatives. The Jacobian of section 8 is computed by analytic differentiation of each invariant with respect to the 126 coordinates — by amputation of one F from the contraction pattern — and not by finite differences. Every row is checked against Euler's homogeneity relation, which gives an independent handle on differentiation errors: a row that fails Euler is wrong regardless of what rank it would have contributed.

Determinants. Where a determinant certifies a rank we compute it twice, by fraction-free Bareiss elimination [29] and by an independent exact routine, and require agreement.

Frozen executions and immutable artifacts. Each rank-matrix cell writes an immutable JSON artifact. Aggregation is a separate, read-only, order-independent step. This separation exists because an earlier incident in this project involved two processes sharing a row cache; the architecture now makes that impossible rather than merely discouraged.

What is prior work. The enumerate–evaluate–relate workflow is due to Elamaran, Ferko and Scarlett [21]. Canonicalisation is `nauty` [22]. Bareiss elimination, CRT and rational reconstruction are classical. What is new here is the certification layer: the exact arithmetic throughout, the explicit minors, the held-out-prime validation, the counterfactual tests, and the discipline that no number reaches the manuscript except through a generated macro.

7 Degree-resolved tensor–spinor equivalence

The two formulations are compared on 78 common sample points at prime 32749, drawn from four families: sparse, structured, generic and a held-out set used only for validation. For each degree we compute the rank of the tensor family, the rank of the spinor family, and the rank of their union. Equality of the first and third means the spinor family adds nothing the tensor family did not already have, and vice versa.

The three ranks are listed degree by degree in table 2 and plotted in figure 1.

degree d	tensor rank	spinor rank	union rank	spans equal
4	1	1	1	yes
6	2	2	2	yes
8	7	7	7	yes
10	14	14	14	yes

Table 2. Degree-resolved ranks. At degree eight the spinor family is the one that includes the tensor words; the narrower family without them is the subject of figure 2.

7.1 Degrees four and six

At degree four the invariant space is one-dimensional and at degree six it is two-dimensional, in both formulations, with the spans equal and the equality confirmed on held-out samples. These degrees are not in doubt and we record them mainly as a check on the machinery.

7.2 Degree eight, and an ablation

Degree eight is where the interesting structure appears. With the full spinor candidate family,

$$\dim A_8^{\text{tensor}} = \dim A_8^{\text{spinor}} = \dim\big(A_8^{\text{tensor}} + A_8^{\text{spinor}}\big) = 7, \tag{7.1}$$

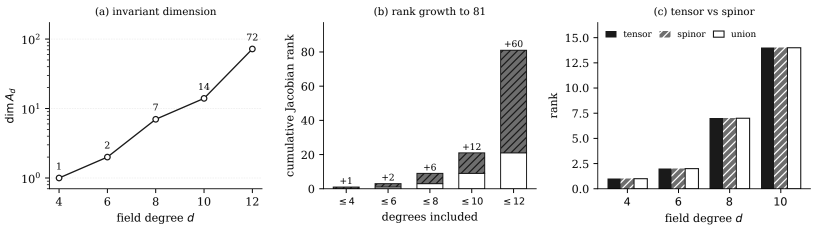

Figure 1. Degree-resolved tensor, spinor and union ranks. At every degree shown the three agree, so neither formulation sees an invariant the other misses.

at all 4 of the 4 primes tested.

The full spinor family includes the tensor words of section 4. Removing them drops the spinor-side rank from 7 to 6, so one invariant direction is lost. The tensor words are therefore not redundant. Figure 2 shows the ablation, and the candidate families it is run over are inventoried in table 3.

candidate family	count
port graphs	70
structured candidates	13
total scheduled	83
total evaluated	83
evaluation errors	0
zero-Jacobian rows	0
Euler homogeneity pass	yes

Table 3. Candidate-family inventory for the generic-rank computation. The schedule is complete: every candidate was evaluated, none errored and none was silently skipped.

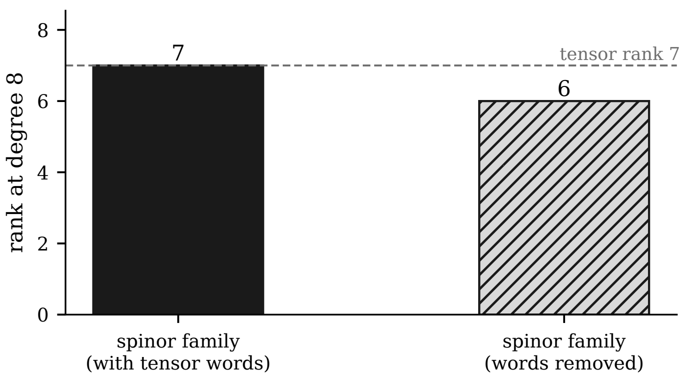

Figure 2. Degree-eight family ablation. Removing the tensor words from the spinor candidate family lowers the rank by one, so within this decomposition they are indispensable.

We state the conclusion in the qualified form the evidence supports: *tensor words are indispensable within the tested candidate-family decomposition.* This is a statement about the families we constructed. It is not a uniqueness theorem, and we do not claim that no other decomposition could reach rank 7 without them.

Remark 3 (Two different degree-eight spinor ranks)*.* The reader comparing against intermediate artifacts of this project will find a degree-eight spinor rank of 6 recorded alongside the 7 quoted here. These are not in conflict: they are the ranks of two different candidate families, before and after the tensor words are included. The ablation above is exactly the difference between them.

7.3 Degree ten

At degree ten both formulations give rank 14, with equal spans and held-out validation passing. This is the atlas dimension $\dim A_{10}$ that the rest of the paper uses.

We should be precise about what identifies this atlas dimension with the full invariant space. The equality $\dim A_{10} = 14$ is exact for the span of the candidates we enumerate; reading it as the dimension of *all* degree-ten Lorentz invariants additionally requires that the enumeration miss nothing. Two independent constructions — contraction graphs and chiral bispinors — were carried out separately and agree here and at degrees four, six and eight, which is meaningful corroboration precisely because a shared omission would have to occur twice in two different languages. It is not a proof of exhaustiveness, and we do not present it as one; the point is recorded again in section 14.

8 Generic functional rank

8.1 The computation

Let $I_1, \ldots, I_N$ be the candidate invariants, $N = 83$, each a polynomial in the $n = 126$ coordinates F^A of Λ^5_+. The generic functional rank of the family is the rank at a generic point of the Jacobian

$$J(F)_i{}^A \;=\; \frac{\partial I_i}{\partial F^A}, \qquad i = 1, \ldots, N, \quad A = 1, \ldots, n, \tag{8.1}$$

an 83×126 matrix. Each row is computed *analytically*, by amputating one factor of F from the contraction pattern that defines I_i, rather than by finite differences; there is no step size and no cancellation error.

Each row carries an independent check. Since I_i is homogeneous of field degree d_i, Euler's relation gives

$$\sum_{A=1}^{n} F^A \, \frac{\partial I_i}{\partial F^A} \;=\; d_i \, I_i, \tag{8.2}$$

which relates a differentiated row to an undifferentiated evaluation and is therefore sensitive to exactly the errors that differentiation introduces. A row failing (8.2) is wrong whatever rank it would have contributed. Every scheduled candidate was evaluated — 83 of 83, with 0 errors and 0 identically-zero rows — and (8.2) holds on every row.

The exact modular rank is 81. Cumulative ranks by degree are 1, 3, 9, 21 and 81 at degrees four, six, eight, ten and twelve respectively. These increments are plotted in figure 1(b). No step in the chain uses floating point and no step uses a tolerance: the candidates are integer-coefficient polynomials, the Jacobian is differentiated analytically by amputation, and the rank is obtained by exact elimination over $\mathbb{F}_p$.

8.2 The certificate

A rank claim backed only by an elimination is a claim about a program. We therefore exhibit an explicit 81×81 minor whose determinant is non-zero, computed by two independent routines that agree. Since a non-vanishing minor over $\mathbb{F}_p$ lifts to a non-vanishing minor over $\mathbb{Q}$ — the determinant is an integer that is non-zero mod p — this gives the characteristic-zero lower bound

$$\text{generic functional rank} \;\geq\; 81 \tag{8.3}$$

with no prime excluded and no appeal to genericity of the sample.

A rank is a number, and on its own it says nothing about *which* invariants realise it. Since the certificate names its own rows, that question can be answered directly: figure 3 sorts the 81 pivot rows by the degree of the candidate and by the family it was constructed in. Two features matter for what follows. The pivots are spread across every degree rather than concentrated at one, so the bound is not an artefact of the degree-twelve block alone; and the tensor words supply pivots of their own, which is the same conclusion the degree-eight ablation of section 7.2 reaches by a different route.

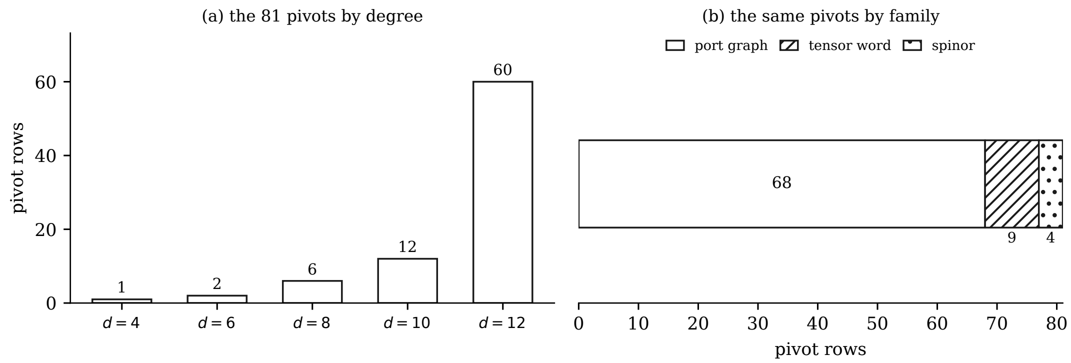

Figure 3. The 81 pivot rows of the certified minor, read from the certificate's own row list. Left, by the field degree of the candidate; right, by the family it belongs to. Neither split is uniform, and neither is concentrated in a single bucket.

8.3 The prime/sample matrix

The computation was repeated across 5 primes and 3 seeds, giving 15 cells, set out in table 4 and figure 4. Every cell returns rank 81, and every cell returns the *same* 81 pivot rows and 81 pivot columns, with 0 unstable.

We are deliberately restrained about what this establishes. Thirteen of the 15 cells were produced by a single run in a separate working tree and were not independently recomputed here, and cells from one run agreeing with each other is close to guaranteed whether or not the run was correct. The agreement across primes and seeds is therefore corroboration, not independent confirmation. What carries weight is narrower and does not depend on the other thirteen: the two cells recomputed in this repository agree with the certificate on total rank, per-degree block ranks, cumulative ranks, pivot rows, pivot columns and pivot row candidate identities; and the lower bound itself rests on a single explicit minor (section 8) that can be checked without reference to the matrix at all.

prime	seed 11	seed 22	seed 33
32693	81	81	81
32713	81	81	81
32717	81	81	81
32719	81	81	81
32749	81	81	81

Two of these cells were recomputed in this repository and agree with the certificate on every recorded quantity. The remaining thirteen were not independently recomputed here; what is established for them is internal consistency plus a shared candidate-ordering hash.

Table 4. The prime/sample rank matrix: 15 cells, every one of rank 81, with identical pivot rows and columns throughout.

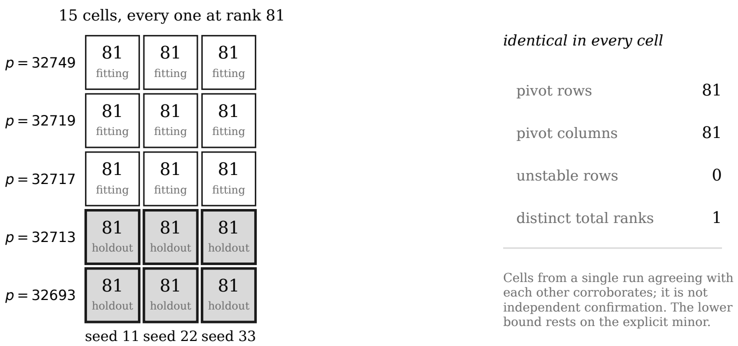

Figure 4. The prime/sample rank matrix. Every cell agrees on rank and on pivot structure.

8.4 What is and is not claimed

The upper bound is analytic and belongs to earlier work [20]; combined with our lower bound it gives 81 exactly. We claim the lower bound and the certificate, not the number.

Two further cautions. First, the statement is about *functional* rank at a generic point: the selected family has generic functional rank 81. We do *not* claim that all 83 candidates are algebraically independent, which is a different and stronger assertion. Second, of the 15 cells, two were recomputed independently in this repository and agree with the certificate on every recorded quantity; the remaining thirteen were not independently recomputed here, and for them what is established is internal consistency together with a shared candidate-ordering hash. This is stated again in the limitations.

9 Reachability of the stress flow at degree ten

9.1 Three spaces, two of which are easily confused

The stress-tensor flow generates new invariants from old by repeatedly applying a construction built from the stress tensor τ of the deformed theory [13, 20]. Schematically, a deformation parameter λ drives

$$\frac{\partial \mathcal{L}}{\partial \lambda} = \mathcal{O}\big[\tau(\mathcal{L})\big], \tag{9.1}$$

where $\mathcal{O}$ is a scalar built from traces of powers of τ. We do not reproduce the construction of [20] here; what the argument below needs from it is only which trace monomials it produces and the field degree at which each begins. Both were determined by instrumenting the closure rather than by reading the construction, so that a generator consumed but not documented would show up. Through degree ten the generators are exactly

$$\begin{array}{lllll} \operatorname{Tr}\tau, & \operatorname{Tr}\tau^2, & \operatorname{Tr}\tau^3, & \operatorname{Tr}\tau^4, & \operatorname{Tr}\tau^5, \\ (\operatorname{Tr}\tau)^2, & \operatorname{Tr}\tau\operatorname{Tr}\tau^2, & \operatorname{Tr}\tau\operatorname{Tr}\tau^3, & (\operatorname{Tr}\tau^2)^2, & \operatorname{Tr}\tau^2\operatorname{Tr}\tau^3. \end{array} \tag{9.2}$$

Write $\deg_F$ for the leading field degree of an expression, that is, the lowest power of F appearing in it. The grading that controls the whole calculation is

$$\deg_F\Big(\operatorname{Tr}\tau^k\Big) = \begin{cases} 4, & k=1, \\ 2k, & k\geq 2, \end{cases} \qquad \deg_F\Bigg(\prod_i \operatorname{Tr}\tau^{k_i}\Bigg) = \sum_i \deg_F\Big(\operatorname{Tr}\tau^{k_i}\Big). \tag{9.3}$$

The exceptional case $k=1$ is the content of theorem 7: the naive count would give $\deg_F \operatorname{Tr}\tau = 2$, and it is the vanishing of the free trace that pushes it to 4. Every entry of (9.3) was checked independently at every prime, with no exceptions.

A target is *generated* when the flow produces it formally, and *activated* at degree d when (9.3) permits it to contribute there. Writing V_d for the span reached at degree d, the closure is the least fixed point of

$$V \longmapsto V + \operatorname{span}\big\{\, g \;:\; g \text{ generated from } V,\ \deg_F g \leq d \,\big\}, \tag{9.4}$$

started from the degree-four seed and iterated until a sweep raises no rank. Termination occurred after 3 sweeps, and over $\mathbb{Q}$ the reachable dimensions by degree are 1, 2, 7 and 11 at degrees four, six, eight and ten.

Two structural facts make the question below well posed. The map induced by (9.1) on each graded piece is linear, and D_{10} is by construction the *span* of the reachable elements; so D_{10} is a linear subspace of A_{10} and the quotient $Q_{10} = A_{10}/D_{10}$ is defined. Asking what the flow reaches at degree ten then requires three spaces to be kept apart.

Definition 4.

- The *full invariant space* A_{10}: all degree-ten Lorentz invariants.
- The *raw target span*: the span of every target the flow generates, *ignoring* whether the flow can actually activate it at this degree.

- The *activated closure* D_{10}: the span of what the flow actually reaches, obtained by iterating activation to a fixed point.

The raw target span has dimension 14 — the same as $\dim A_{10}$. If one identifies it with D_{10}, the quotient comes out zero and the conclusion is that the flow reaches everything. That identification is false, and this project made it before catching it. In symbols, writing $\mathcal{T}$ for the set of targets the flow generates and $\mathcal{A}$ for the activation rule of section 10,

$$\operatorname{span}\mathcal{T} \;=\; A_{10} \qquad \text{but} \qquad D_{10} \;=\; \operatorname{span}\left(\bigcup_{k\geq 0} \mathcal{A}^k(\mathcal{T}_0)\right) \;\subsetneq\; A_{10}, \tag{9.5}$$

where $\mathcal{T}_0$ is the seed. The left equality is why the conflation is tempting — the raw span really does exhaust A_{10} — and the strict inclusion on the right is the entire content of theorem 5. The raw span is retained in the test suite as a negative fixture so the error cannot recur silently.

9.2 The exact rational computation

The closure is a fixed-point computation, and running it modulo p would leave the answer exposed to exceptional primes in precisely the place where no spanning-set argument is available. We therefore ran it over $\mathbb{Q}$.

Of the closure's targets, 49 were already integral. The remaining 9 were lifted by CRT across 5 fitting primes followed by rational reconstruction, and the lift was validated at the held-out prime 32771, with no mismatches. The closure was then iterated in exact $\mathbb{Q}$ arithmetic and reached its fixed point after 3 sweeps.

It is worth showing why this was necessary rather than merely careful, because the cost of exact arithmetic is real and a reader is entitled to ask whether one modular solve would have done. Figure 5 plots the size of the largest coefficient the closure actually produces at each degree, taken from its own exact output. Through degree eight the coordinates are small integers and nothing is at risk. At degree ten one basis vector of the reachable space carries a numerator of twenty-nine bits over a five-digit denominator — an order of magnitude beyond what a single prime near 2^{15} can represent. Read modulo one such prime that coefficient is indistinguishable from a small residue, and the fixed point would have been reported at the wrong dimension with nothing in the output to indicate it. The lift is what makes the degree-ten row recoverable at all.

Theorem 5 (Exact degree-ten reachability)**.** *Over the rationals,*

$$\dim_{\mathbb{Q}} A_{10} = 14, \qquad \dim_{\mathbb{Q}} D_{10} = 11, \qquad \dim_{\mathbb{Q}} Q_{10} \;=\; 3. \tag{9.6}$$

The last equality deserves a word, because it is stronger than the modular computation it replaced. Running the closure over $\mathbb{Q}$ removes the modular admission test that made the earlier figure a lower bound. At the fixed point no further generated direction raises the rank over $\mathbb{Q}$, so the reached span *is* D_{10} and its rank is an equality, not a bound. No separate upper-bound argument is required. Table 5 lists the degree-ten invariants and the block each occupies.

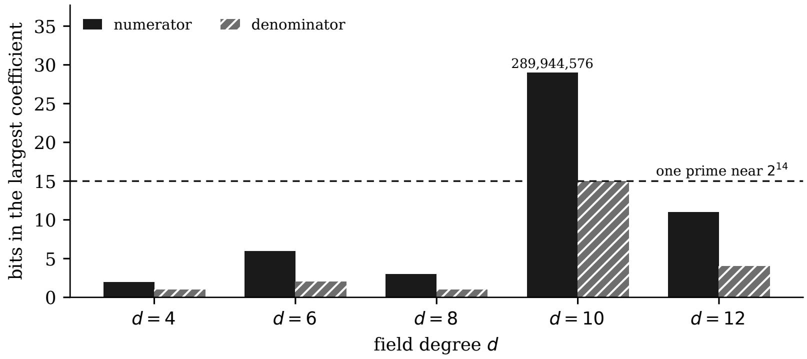

Figure 5. Size of the largest numerator and denominator appearing in a basis vector of the flow-reachable space, by degree, in bits. The dashed line is the representable range of a single prime near 2^{15}. Degree ten is the first place the exact coefficients leave that range, which is why that row is lifted by the Chinese remainder theorem and rational reconstruction rather than read off one modular solve.

space	$\dim_{\mathbb{Q}}$	how established
A_{10}	14	atlas, spanning-set bound
raw target span	14	all generated targets, kept as a negative fixture
D_{10}	11	exact rational closure, fixed point
Q_{10}	3	quotient

Table 5. Degree-ten dimensions. The closure reached its fixed point after 3 sweeps in exact rational arithmetic.

The lower-bound side is certified by an explicit 11×11 integer minor with non-zero determinant, computed by Bareiss elimination and confirmed by an independent exact rational verifier. Three exact annihilating covectors exhibit the quotient directly, and the quotient representatives are indexed by the free columns not occupied by a pivot.

representative	origin
q_1	free column 5
q_2	free column 6
q_3	free column 11

Table 6. Quotient representatives. The free columns are those not occupied by a pivot of the rank-11 minor certifying $\dim_{\mathbb{Q}} D_{10}$.

9.3 Reading the result

There are 3 independent degree-ten invariant directions that the stress flow does not reach. The subspaces of A_{10} that this paper determines exactly stand in the relations

$$D_{10} \subsetneq A_{10}, \qquad \dim_{\mathbb{Q}} A_{10}/D_{10} = 3, \qquad \dim_{\mathbb{Q}}(B_{10} \cap P_{10}) = 1, \qquad \dim_{\mathbb{Q}}(B_{10} + P_{10}) = 13, \tag{9.7}$$

with $\dim_{\mathbb{Q}} B_{10} = 12$ and $\dim_{\mathbb{Q}} P_{10} = 2$; the last two are established in section 11. Every dimension in (9.7) is exact over $\mathbb{Q}$, not modulo a prime.

9.4 How many directions are needed, in any basis

Proposition 6 (Cardinality minimality). *If $S \subset A_{10}$ is any set with $D_{10} + \operatorname{span}(S) = A_{10}$, then $|S| \geq \dim Q_{10}$.*

Proof. Apply the quotient map $\pi : A_{10} \to Q_{10}$. From $D_{10} + \operatorname{span}(S) = A_{10}$ we get $\operatorname{span}(\pi(S)) = Q_{10}$, and a span of $|S|$ vectors has dimension at most $|S|$. □

This is elementary linear algebra and we claim no novelty for it. We state it because it converts a basis-dependent assertion into a basis-free one: no closing set of any kind, in any basis, has fewer than 3 elements. What it does *not* give is removal minimality of the particular set we exhibit, which we have checked only in the fixed graph basis and under the permutation subgroup, and which remains open in general (section 14).

10 The free stress-tensor trace and activation

The leading-degree bookkeeping of the closure rests on one physical input: the degree at which $\mathrm{Tr}(\tau)$ first contributes. If it began at degree two rather than four, every generated target would land in the wrong graded piece and theorem 5 would fall with it. This section derives the input.

Theorem 7 (Vanishing of the quadratic trace). *For a self-dual five-form in ten dimensions, the free stress tensor has vanishing trace, so $\mathrm{Tr}(\tau)$ first contributes at field degree four.*

Proof. Write the free stress tensor of a p-form field strength in d dimensions in its general improved form,

$$T_{\mu\nu} \;=\; \frac{1}{(p-1)!}\, F_{\mu\alpha_2\ldots\alpha_p}\, F_\nu{}^{\alpha_2\ldots\alpha_p} \;-\; c\,\eta_{\mu\nu}\,\langle F, F\rangle, \qquad \langle F, G\rangle \;=\; F_{\mu_1\ldots\mu_p} G^{\mu_1\ldots\mu_p}, \tag{10.1}$$

with c an improvement coefficient left free. Contracting with $\eta^{\mu\nu}$, the first term returns $\langle F, F\rangle$ up to the normalisation carried by the factorial and the second returns $-\,c\,d\,\langle F, F\rangle$, so

$$T^\mu{}_\mu \;=\; (1 - c\,d)\,\langle F, F\rangle. \tag{10.2}$$

Whatever c may be, the trace is a multiple of the single scalar $\langle F, F\rangle$; no other quadratic Lorentz invariant exists, since the only way to contract two copies of a p-form into a scalar is the full contraction. The conclusion below is therefore independent of the improvement term and of the overall normalisation.

It remains to show that this one scalar vanishes. Specialise to $p = 5$, $d = 10$, so that F is a middle-degree form of *odd* degree in an even-dimensional space, and consider the top form $F \wedge F$. Interchanging two factors of a q-form costs $(-1)^{q^2} = (-1)^q$, which for $q = 5$ is -1; hence

$$F \wedge F \;=\; -\,F \wedge F \qquad \Longrightarrow \qquad F \wedge F \;\equiv\; 0 \tag{10.3}$$

identically, for every five-form whatsoever. Since $\langle F, \star F\rangle\,\mathrm{vol} \propto F \wedge F$, the pairing $\langle F, \star F\rangle$ vanishes as well. On either eigenspace of the Hodge star we have $\star F = \pm F$, and therefore

$$\langle F, F\rangle \;=\; \pm\,\langle F, \star F\rangle \;=\; 0. \tag{10.4}$$

Substituting (10.4) into (10.2) gives $T^\mu{}_\mu = 0$ for every c. □

Two features of this argument are worth isolating, because both are places where a plausible-looking variant of the calculation gives a different answer.

The vanishing in (10.3) is a statement about form parity alone and holds for *any* five-form, self-dual or not; self-duality enters only at (10.4), to convert $\langle F, \star F \rangle$ into $\langle F, F \rangle$. The two steps are independent, and a test suite that checked only self-dual samples would confirm the conclusion without ever exercising the mechanism behind it. Ours checks both eigenspaces — the argument predicts vanishing on each — and, as a control, checks that $\langle F, F \rangle \neq 0$ on an *unprojected* five-form. Without that control the first three checks would be satisfied vacuously by a quantity that vanished for trivial reasons.

The sign $\star^2 = +1$ used in (10.4) is signature-dependent. In Euclidean signature $\star^2 = -1$ on middle-degree forms in ten dimensions, the eigenvalues are $\pm i$, and there is no real eigenspace decomposition to speak of. This is not a hypothetical: an earlier version of the control called a projector routine with an argument believed to select the anti-self-dual space when it in fact selected Euclidean signature, and the resulting non-vanishing was briefly read as signal. The value of $\star^2$ is now asserted by test wherever the decomposition is used rather than presumed.

The argument is short, and its shortness is the point: it uses no property of the particular theory beyond self-duality and the parity of the form degree, so it survives any reformulation of the free dynamics.

Remark 8 (Free versus deformed trace). Theorem 7 is about the *free* stress tensor, while the flow (9.1) involves the deformed τ. This is the right object for the purpose. The grading argument needs only the field degree at which $\mathrm{Tr}(\tau)$ *begins*, and the leading term of the deformed trace is the free trace: the deformation contributes corrections of strictly higher degree. We make no claim that the deformed trace vanishes — in general it does not, which is precisely what makes the flow non-trivial.

Remark 9 (Which stress tensor, and why this is a review item). The theorem establishes tracelessness for a stress tensor of the standard p-form shape with an arbitrary improvement term. Whether the object appearing in the flow of [20] is of that shape, in the formulation the authors intend, is a question about their conventions rather than about our computation, and we have not been able to settle it from the published text alone. Because every degree-ten result depends on it, we flag it explicitly as review item **G-10** rather than assuming the favourable reading. If the intended object is of a different shape, the improvement-independence above means the conclusion is likely to survive, but "likely" is not the standard the rest of this paper is held to.

Verification, with a control. The computational half is checked by code that imports no flow machinery, re-deriving the statement from the five-form and the Hodge star alone. The vanishing is confirmed on both eigenspaces at four primes. Crucially the suite also contains a *control*: $\langle F, F \rangle$ must *not* vanish on an unprojected five-form. Without that control the vanishing tests would prove nothing, since a function that vanished identically would pass them all.

The counterfactual. To show the derivation is load-bearing rather than decorative we forced the opposite assumption — $\mathrm{Tr}(\tau)$ beginning at degree two — and re-ran the closure. The result is $Q_{10} = 0$, against 3 as derived, with the degree-ten reachable dimension rising to 14 and swallowing the whole atlas. The counterfactual reproduces exactly the wrong answer this project obtained at an earlier stage, and it is retained as a regression fixture.

suite	count / effect
tensor suite	254
bridge suite	86
manuscript claim gates	72
G-10 counterfactual	$Q_{10} \to 0$
orientation branch flip	breaks a working prime

Table 7. Mutation and regression coverage. The last two rows are counterfactual tests: each deliberately breaks an assumption and checks that the stated consequence follows.

11 The published basis and the product sector

Let B_{10} denote the span of the degree-ten structures given in [19], and P_{10} the product subspace. Both are spanned by explicit elements, so their rational dimensions follow from modular ranks with no prime excluded: $\dim_{\mathbb{Q}} B_{10} = 12$ and $\dim_{\mathbb{Q}} P_{10} = 2$.

The interesting quantity is the intersection. To settle it over $\mathbb{Q}$ we lifted all 12 published coordinate vectors by CRT across 7 fitting primes and rational reconstruction, validated at the held-out prime 32783 with no mismatches, and computed the intersection exactly.

Theorem 10. $\dim_{\mathbb{Q}}(B_{10} \cap P_{10}) = 1$, *and* $\dim_{\mathbb{Q}}(B_{10} + P_{10}) = 13$.

The one-dimensional intersection can be written out. With $I^{(k)}$ denoting the published structures in the numbering of [19] and $I_1^{(4)} I_1^{(6)}$ the product of the degree-four and degree-six invariants,

$$-55\, I^{(1)} - 252\, I^{(2)} + 4200\, I^{(5)} + 5040\, I^{(6)} \;=\; -1040\, I_1^{(4)} I_1^{(6)}. \tag{11.1}$$

This identity was verified at the fresh prime 32869 on 6 independently drawn samples, none of which were used in the reconstruction.

Because (11.1) carries specific integer coefficients, it is not invariant under rescaling the objects entering it, unlike every rank and dimension elsewhere in this paper. The coefficients are stated relative to the normalisation of the published structures in [19] and of the degree-four and degree-six invariants fixed in appendix D. Under a rescaling $I^{(k)} \to c_k I^{(k)}$ the coefficients transform by the inverse factors; what is normalisation-independent is the statement that the intersection is one-dimensional and that its generator is supported on the four published structures listed. The published basis is given in table 8 and the product sector in table 10.

quantity	value
published structures lifted	12 of 12
fitting primes	7
held-out prime	32783
held-out mismatches	0
fresh-sample prime	32869
fresh samples	6
$\dim_{\mathbb{Q}}(B_{10} \cap P_{10})$	1

Table 8. Reconstruction status for the published span. Every published structure lifted to $\mathbb{Q}$, and no held-out prime disagreed.

quantity	value
$\dim(B_{10} \cap P_{10})$	1
$\dim(B_{10} + P_{10})$	13
non-product content of B_{10}	11
$\dim(B_{10} \cap G_{10})$	10
$\dim(P_{10} \cap G_{10})$	0
$\dim(D_{10} \cap B_{10})$	9

Table 9. Span comparisons among the published, graph-generator, product and reachable subspaces.

Consequences. The published span contains exactly one product direction, so its non-product content is 11-dimensional. In particular B_{10} is not a complement to the products. This is a structural fact about the published choice of structures, not a criticism of it: nothing in [19] claims otherwise, and the structures there are presented as explicit expressions rather than as a basis with specified relations.

Remark 11 (A check that failed first). The verification of (11.1) initially failed at a fresh prime, on all six samples. The cause was in the check, not the identity: the two sides were being reduced by their greatest common divisors separately, which destroys the relative scale that the identity is precisely about. Recomputing on a single common scale, it holds. We record this because a check that fails and is then quietly adjusted until it passes is worth nothing, and the only defence against that is to say what was adjusted and why.

space	dimension over $\mathbb{Q}$
$\dim_{\mathbb{Q}} A_{10}$	14
$\dim_{\mathbb{Q}} G_{10}$	12
$\dim_{\mathbb{Q}} P_{10}$	2
$\dim_{\mathbb{Q}} B_{10}$	12
$\dim_{\mathbb{Q}} D_{10}$	11
$\dim_{\mathbb{Q}} Q_{10}$	3

Table 10. Product and primitive dimensions at degree ten, all established exactly over the rationals.

12 Reproducibility and proof architecture

The results above are computational, and a computational result that cannot be re-run is an assertion. The architecture supporting them has the following components.

Test suites. 254 tests on the tensor side and 86 on the bridge side, run from a clean clone.

Counterfactual and mutation tests. Several tests deliberately break an assumption and assert the consequence: the G-10 counterfactual (section 10), the orientation branch flip (section 5.3), and the raw-target negative fixture (section 9). These matter more than the passing tests, because a suite that only confirms what the code already does cannot detect a shared assumption that is wrong.

Generated macros. Every scientific number in this document is a macro expanded from `generated/numbers.tex`, which is produced from result artifacts. Typing a number into the prose is prevented by convention and checked by the claim gates.

Claim gates. A set of automated checks run against the compiled source, each testing a specific wording or consistency requirement — that no unscoped completeness or canonicality claim appears, that a quantity certified as a bound is never described as an equality, that documentation does not contradict the artifacts. Each gate was negative-tested: deliberately violated, to confirm it fails.

Immutable artifacts and deterministic aggregation. Cells are written once and aggregated read-only, with the aggregation independent of order.

Deterministic figures and byte-reproducible archives. Figures are generated from artifacts with timestamps suppressed, so repeated runs produce byte-identical output. The tests run along that route are summarised in table 7, and the route itself is short enough to state in a sentence: from a clean clone, the suites run, the certificate scripts regenerate their artifacts, the figures and tables are rebuilt from those artifacts, the macros are extracted, the manuscript compiles, and every gate must pass or the build fails.

Table 11 lists every certificate the draft rests on, with the script that produced it, and table 12 records the environment the certified runs were made in.

13 Physical interpretation

We are deliberately cautious here, and we separate what has been established from what it might mean.

What has been established is a statement about a construction: the stress-tensor flow, as formulated in [20] and in the conventions fixed in appendix A, does not reach 3 independent directions of the degree-ten invariant space. That is a fact about the flow, not about nature.

It is nonetheless a suggestive fact, because the analogous question in six dimensions appears to have a different answer. There the chiral two-form theories and their $T\bar{T}$-like

claim	proof type	instrument
$\dim_{\mathbb{Q}} D_{10} = 11$	exact rational certificate	fixed point + 11×11 minor
$\dim_{\mathbb{Q}} Q_{10} = 3$	exact rational certificate	quotient of the above
$\dim_{\mathbb{Q}}(B_{10} \cap P_{10}) = 1$	exact rational certificate	CRT lift + explicit generator
generic functional rank ≥ 81	exact modular certificate	81×81 minor, two routines
generic functional rank $= 81$	analytic, cited	upper bound from the literature
G-10 trace vanishing	analytic theorem	improvement-independent, with control
bridge rank 126	exact modular certificate	left inverse and round trip

Table 11. Proof certificates for the central claims. The distinction between the exact lower bound and the cited analytic upper bound on the generic rank is maintained throughout.

component	version
Python	3.13.13
NumPy	2.5.1
pynauty	2.8.8.1
opt_einsum	3.4.0
commit	`c213c7404bfb`

Table 12. Reproduction environment. `opt_einsum` is required, not optional: it supplies the contraction order and the intermediate-size and flop budgets the modular contractor enforces. Without it those budgets are not checked at all and the degree-eight evaluator is terminated by the kernel rather than raising. Computed values are unaffected.

flows are reported to exhibit a universality [13, 14]. We take that characterisation from those works; we have not verified their results and the comparison is theirs, not ours. Read that way, the ten-dimensional degree-ten sector may be the first place in this programme where the flow provably fails to be exhaustive — but establishing that it is the first would require a survey we have not done. Whether the missed directions correspond to genuinely inaccessible deformations, or merely to deformations this particular flow does not generate, is not settled by anything here.

The trace structure of section 10 is a conformal statement in the free theory and it is standard; we use it as an input to the grading, not as a result.

Type IIB, as motivation only. This paragraph is context for why the problem is worth attention; it establishes nothing and no later statement depends on it. The self-dual five-form is the Ramond–Ramond field strength of Type IIB supergravity, whose covariant field equations were given by Schwarz [30] and for which manifestly Lorentz-invariant actions were later constructed by Dall'Agata, Lechner and Tonin [31] using precisely the chiral-boson machinery referred to above. Higher-derivative corrections involving the five-form have been studied directly [32], within the broader eight-derivative programme [33, 34], and non-perturbatively through D-instanton contributions to the R^4 sector [35]. It is natural to ask whether the degree-ten invariants classified here constrain those corrections. We do not answer that question and we make no Type IIB claim. The obstacles are concrete: the corrections live in a supersymmetric theory with additional fields, the relevant invariants are not obviously the Lorentz invariants of a free five-form in isolation, and the action formulation carries the well-known difficulties [1, 6]. We record Type IIB as context and as motivation, and nothing in this paper should be read as a prediction about it.

What is explicitly not claimed. No Type IIB effective-action prediction. No statement about causal propagation or hyperbolicity of any deformed theory. No supersymmetry statement. No all-order theorem about the flow.

14 Limitations and open problems

1. **Generator extension.** We compute what the flow reaches from the generators it is given. Whether an enlarged generator set reaches more of A_{10} is not addressed.

2. **Removal minimality.** Proposition 6 gives cardinality minimality in any basis. Removal minimality of the particular closing set we exhibit — that no element of it can be dropped — has been checked only in the fixed graph basis and under the permutation subgroup. Minimality under arbitrary GL change of basis would require regenerating the certificates in the new basis and has not been attempted.

3. **No canonicality.** We exhibit bases; we do not claim any of them is canonical or unique, and no such claim should be read into the word "basis".

4. **Enumeration exhaustiveness is corroborated, not proved.** Every dimension in this paper is exact for the span of the candidate family we enumerate. Identifying that span with the space of *all* degree-ten Lorentz invariants requires the enumeration to be exhaustive, and we do not prove that it is. What we have is two independently implemented constructions, contraction graphs and chiral bispinors, whose spans agree at degrees four, six, eight and ten; a missing invariant would have to have been missed twice, in two different languages. That is corroboration and it is not a theorem. A proof would need a Hilbert-series or character-theoretic count of the degree-ten invariants of the 126-dimensional module, which we have not carried out and which is the single most valuable check anyone could add to this work.

5. **Degree twelve.** Degree twelve supplies partial certified input to the generic rank calculation and is excluded from this paper's tensor–spinor equivalence and degree-ten classification claims. We do not claim a complete degree-twelve equivalence.

6. **No invariant-ring presentation.** We give dimensions and bases of graded pieces, not generators and relations for the full invariant ring, and no Hilbert series is claimed as a result here.

7. **No all-order flow theorem.** Everything about reachability is stated at degree ten. No induction to all orders is offered.

8. **Rank-matrix independence.** Two of the 15 cells were independently recomputed in this repository. The other thirteen carry internal consistency and a shared ordering hash, which is weaker.

9. **A source ambiguity, avoided rather than resolved.** Two of the published degree-ten expressions are ambiguous as printed, because bracket colour — which fixes the order of operations — does not survive text extraction. We implemented both readings and report the consequences of each, and the compact Q_{10} basis is chosen to avoid the ambiguous candidates entirely, so no result here depends on the resolution. The ambiguity is avoided, not resolved; confirming the source's intent requires its authors.

10. **Physical interpretation.** As set out in section 13, the reachability result is a statement about a construction. Its physical significance is open.

11. **"Basis" is used loosely.** We exhibit spanning sets of the stated dimension and call them bases. Since removal minimality is open in general (item 2), a reader should read "basis" throughout as "spanning set of cardinality equal to the dimension", which is what is certified.

12. **No Hilbert-series cross-check.** The natural independent check on graded dimensions would be a Hilbert series for the invariant ring. We did not compute one. Its absence is a missing corroboration rather than a defect, but a reader entitled to expect it should know it is not here.

13. **Held-out primes.** The D_{10} lift was validated at one held-out prime; the B_{10} lift at one held-out prime and again on fresh samples at a further prime. A single held-out prime leaves a probability of order $1/p$ per row of a silent false pass. That is small, and it is not zero.

14. **Independence of the two determinant routines.** The Bareiss and exact-rational routines are separate implementations but share this project's modular-arithmetic layer. A defect confined to that layer would not be caught by their agreement.

15. **The macro-generation link.** Every number in this paper is generated from an artifact by one script. A defect in that script would propagate to the prose and to the claim gates alike, since both read its output. The mitigation is that the gates compare macros against *independently parsed* artifact fields rather than against each other, but the link is a single point of failure and we name it as one.

16. **Optional-dependency code path.** The certified runs used the built-in contraction ordering because `opt_einsum` was not installed. We expect the accelerated path to give identical values, since contraction order affects speed and not arithmetic, but we did not run the full certification both ways, so this is an expectation rather than a verified claim.

Table 13 classifies every claim in the paper by the strength of the evidence behind it, from theorem to expectation.

classification	meaning
analytic theorem	proved in the text; no computation relied upon
exact rational certificate	established over $\mathbb{Q}$; no prime excluded
exact finite-field certificate	exact mod p; a spanning-set or one-sided argument carries it to $\mathbb{Q}$
multi-prime validation	agreement across primes and seeds; evidence, not proof
limitation	stated scope restriction
open problem	not established here

Table 13. Claim-strength classification used in the claim ledger and throughout the text.

15 Conclusions

We have determined the degree-ten Lorentz-invariant structure of a self-dual five-form in ten dimensions, exactly and with certificates.

The full degree-ten invariant space has dimension 14 over $\mathbb{Q}$. The subspace reachable by the stress-tensor flow has dimension 11, and the quotient therefore has dimension 3: three independent directions the flow does not reach. Both dimensions are established over the rationals, not merely modulo a prime, by lifting the closure's targets and running it to its fixed point in exact rational arithmetic. The span of the published degree-ten structures meets the product sector in exactly one dimension, and that intersection is written out as an explicit integer identity, equation (11.1).

Supporting these are an exact tensor–spinor bridge of certified rank 126 with an exact left inverse, a degree-resolved verification that the two formulations span the same invariants at degrees four, six, eight and ten, and an exact lower bound of 81 on the generic functional rank of the candidate family, certified by an explicit 81×81 minor. The matching upper bound is analytic and is due to earlier work.

Along the way two defects in our own earlier treatment were found and corrected: the conflation of the raw target span with the flow-reachable subspace, which had produced a quotient dimension of zero, and an unpinned square-root branch in the frame construction, which had left the self-dual and anti-self-dual channels to be selected by accident. Both are now covered by tests that fail if the defect returns.

The open questions we regard as most interesting are whether the three missed directions have a physical meaning, and whether an enlarged generator set would reach them.

AI-assisted preparation

[Draft statement pending author review.] This work was prepared with substantial assistance from an AI coding assistant. Being specific is more useful than being reassuring, so: the assistant generated and refactored a large fraction of the source code; it debugged failures, including several of the defects reported in this paper; it assisted with literature search and with the drafting of this manuscript; it generated tests, including the counterfactual and control tests; and it operated the tools that executed the verification scripts, so the verification runs reported here were AI-operated. It is therefore not the case that the computations were verified by a human independently of the system that produced them; what a human did was specify the problems, choose the conventions, decide which claims to make and at what strength, and take responsibility for the result. No AI system is an author, and none is listed as one. Final responsibility for every claim rests with the human authors, once determined.

Data and code availability

[Draft statement pending authorship, licence and archival decisions.] The source code, result artifacts and certificates supporting this paper are in the public repository

`https://github.com/uskutsav-cpu/selfdual-5form-invariants`

on the branch `publication/jhep-mentor-draft`. The exact commit is recorded in the build manifest accompanying this draft.

The following remain unresolved and are flagged rather than answered:

- **[No software licence has been selected.]** Until one is, the code is public but not licensed for reuse, and that is a real limitation on reproducibility rather than a formality.
- **[No DOI has been minted]** and no archival snapshot has been deposited.
- The mentor-review correspondence and review archive are deliberately excluded from the public repository.

Acknowledgments

[Pending author decision.]

A Conventions

Everything in this paper depends on a consistent set of conventions, and several of the results are sensitive to sign choices in a way that is easy to underestimate. They are collected here in full.

Metric and signature. Spacetime is $\mathbb{R}^{1,9}$ with $\eta_{\mu\nu} = \mathrm{diag}(-1, +1, +1, \ldots, +1)$, indices $\mu = 0, 1, \ldots, 9$. Indices are raised and lowered with η. The split signature $(5, 5)$ appears only as the natural home of the oscillator frame in appendix B; it is a different real form of the same complex algebra, not a change of frame within the Lorentzian one.

Orientation and epsilon. The orientation is fixed by $\epsilon_{01\ldots9} = +1$, hence $\epsilon^{01\ldots9} = -1$ in Lorentzian signature. The volume form is $\mathrm{vol} = \frac{1}{10!}\, \epsilon_{\mu_1\ldots\mu_{10}}\, dx^{\mu_1} \wedge \cdots \wedge dx^{\mu_{10}}$.

Hodge star and self-duality. On five-forms,

$$(\star F)_{\mu_1\ldots\mu_5} = \frac{1}{5!}\,\epsilon_{\mu_1\ldots\mu_5}{}^{\nu_1\ldots\nu_5} F_{\nu_1\ldots\nu_5}, \tag{A.1}$$

and with the above conventions $\star^2 = +1$ on middle-degree forms in ten Lorentzian dimensions. This sign is what makes the eigenspaces real; it is verified computationally rather than assumed, and a test asserts it directly.

The sign is signature-dependent: in Euclidean signature $\star^2 = -1$ on middle-degree forms in ten dimensions, the eigenvalues are $\pm i$, and there is no real eigenspace decomposition at all. We state this here, where the convention is fixed, rather than only in passing, because this project once mistook a Euclidean projector for an anti-self-dual one and read the resulting non-vanishing as a signal (appendix I). Any routine taking a signature flag must therefore be called deliberately, and the value of $\star^2$ is checked rather than presumed wherever the decomposition is used. A five-form is self-dual when $\star F = +F$ and anti-self-dual when $\star F = -F$; both eigenspaces have dimension 126.

Inner product. $\langle F, G\rangle = F_{\mu_1\ldots\mu_5} G^{\mu_1\cdots\mu_5}$, with all five indices raised by η. This is the pairing used throughout appendix I.

Clifford algebra. $\{\Gamma_\mu, \Gamma_\nu\} = 2\eta_{\mu\nu}\mathbf{1}$, with 32×32 gamma matrices. Antisymmetrised products are written $\Gamma^{\mu_1\cdots\mu_k} = \Gamma^{[\mu_1}\cdots\Gamma^{\mu_k]}$ with unit weight, so that $\Gamma^{\mu_1\cdots\mu_k}$ agrees with $\Gamma^{\mu_1}\cdots\Gamma^{\mu_k}$ when all indices are distinct.

Chirality. $\Gamma_{11} = \Gamma^0\Gamma^1\cdots\Gamma^9$ satisfies $\Gamma_{11}^2 = +1$, and the chiral projectors are $P_\pm = \frac{1}{2}(1 \pm \Gamma_{11})$. The chiral module S_+ has dimension 16.

Charge conjugation. C satisfies $C\Gamma^\mu C^{-1} = -(\Gamma^\mu)^T$ in the convention used here. The relevant consequence is the symmetry of $C\Gamma^{(5)}$ on the chiral subspace, which is what allows the bridge (5.1) to land in $\mathrm{Sym}^2 S_+$. This symmetry is checked directly in the implementation.

Five-gamma duality. On the chiral subspace $\Gamma^{(5)}$ is not independent of its dual: the self-dual part of $\Gamma^{(5)}$ acts non-trivially and the anti-self-dual part annihilates S_+. This is the representation-theoretic reason the bridge sees only one eigenspace, and it is the fact that the orientation branch of appendix B can silently invert.

Normalisation. The $1/5!$ in (5.1) is a choice. Every dimension, rank and intersection reported in this paper is invariant under rescaling the bridge, so nothing depends on it. It is fixed only so that the round-trip identity $\Phi^-\Phi = \mathrm{id}$ holds on the nose rather than up to a constant.

Field degree. "Degree" always means degree in F, never degree as a differential form. An invariant of field degree d is a polynomial of degree d in the 126 components of F. Only even field degrees occur.

B The orientation-fixed finite-field bridge

This appendix records the construction of the null frame over $\mathbb{F}_p$, the square-root branch that it leaves free, and the regression that exposed the consequences of leaving it free.

The frame congruence. The oscillator basis diagonalises the metric into 5 hyperbolic pairs. Passing to it requires a matrix L with

$$L^T \eta L = \eta_{\text{null}}, \tag{B.1}$$

where η_{null} is the off-diagonal pairing of the null frame. Over $\mathbb{R}$ one fixes L by continuity from the identity. Over $\mathbb{F}_p$ there is no such notion, and (B.1) determines L only up to the choices made when extracting square roots.

The determinant character. Any two solutions of (B.1) differ by an element of the orthogonal group of η, which has two components. The component is detected by $\det L$, and the two branches of the square root land in different components. This is the entire mechanism: the branch choice is an orientation choice.

How the orientation propagates. An orientation reversal flips the sign of ϵ, hence of $\star$, hence exchanges the $+1$ and -1 eigenspaces. The gamma map of appendix C annihilates one eigenspace and is injective on the other. Composing an orientation-reversed frame with the same gamma map therefore yields the projector onto the *anti*-self-dual space while every variable in the program is still named as though it were the self-dual one. Nothing raises an error; the first symptom is a downstream rank of zero where 126 was expected.

The $p = 32707$ regression. Three cells of the rank matrix failed at $p = 32707$, at every seed. Two observations show the prime is not exceptional:

1. Flipping the branch by hand at $p = 32707$ restores the expected rank.
2. Flipping the branch by hand at a prime that had been working breaks that prime in exactly the same way.

The failure therefore tracks the branch, not the prime, and excluding the prime would have concealed a defect that was live at every prime and merely latent at most of them.

The fix. The frame routine now enumerates both branches, constructs the bridge with each, and returns the one under which the self-dual space is the surviving one — that is, the branch for which the composite has rank 126. If neither branch works it raises rather than returning a guess. The selection is deterministic and depends only on p.

Independent verification. A dedicated test suite covers the fix. Beyond checking that all primes now behave identically, it contains a test that deliberately selects the other branch and asserts that the construction breaks. Without that test a future refactor could drop the normalisation and the suite would still pass at most primes, which is precisely the situation that produced the defect in the first place.

Effect on published values. None. Every certificate had been computed at a prime where the unpinned code already selected correctly, and recomputation after the fix returned identical values. The change is that the convention now holds by construction.

C The exact bridge and its inverse

Forward map. With conventions as in appendix A,

$$\Phi(F)_{ab} = \frac{1}{5!}\, F_{\mu_1\cdots\mu_5}\, (C\Gamma^{\mu_1\cdots\mu_5})_{ab}, \qquad a, b = 1, \ldots, 16. \tag{C.1}$$

In the implementation F is carried as a vector of 126 independent components in a fixed ordering of index five-tuples, and Φ is a 136×126 integer matrix reduced mod p.

Gamma trace. The gamma trace $(\gamma\text{-tr}\,\psi)_\mu = (\Gamma^\nu)^{ab}\psi_{ab}\,\eta_{\mu\nu}$-type contraction has rank 10 on $\mathrm{Sym}^2 S_+$. Hence

$$\dim\ker(\gamma\text{-tr}) = 136 - 10 = 126 = \dim \Lambda^5_+. \tag{C.2}$$

The certified statement is that $\operatorname{im}\Phi \subseteq \ker(\gamma\text{-tr})$ and that the inclusion is an equality by rank.

Rank and kernel. $\operatorname{rank}\Phi = 126$ on Λ^5_+, and $\ker\Phi \cap \Lambda^5_+ = 0$. Both are computed by exact elimination over $\mathbb{F}_p$ at each prime used. On Λ^5_- the map vanishes identically, which is the statement that appendix B shows can silently invert.

Left inverse and round trip. Because Φ has full column rank, a left inverse Φ^- exists over $\mathbb{F}_p$ and is constructed explicitly by exact elimination. The certified identity is

$$\Phi^-\Phi = \mathrm{id}_{126}, \tag{C.3}$$

checked as a matrix identity, not merely on sampled vectors. Note that $\Phi\Phi^-$ is *not* the identity on $\mathrm{Sym}^2 S_+$; it is the projector onto the gamma-traceless subspace, which is the correct expectation and is checked as such.

Equivariance. For Λ in the Lorentz group acting on the tensor side and $S(\Lambda)$ the corresponding spinor representative, the certified statement is

$$\Phi(\Lambda \cdot F) = \chi(\Lambda)\, S(\Lambda)\, \Phi(F)\, S(\Lambda)^T \tag{C.4}$$

where the character χ is solved for rather than assumed. In practice $\chi \equiv 1$ in the normalisation of appendix A; solving for it rather than assuming it is what turns the equivariance check into a test that can fail.

Normalisation independence. Rescaling $\Phi \to \lambda\Phi$ rescales $\Phi^- \to \lambda^{-1}\Phi^-$ and leaves every rank, dimension and intersection in this paper unchanged. The normalisation is therefore fixed for tidiness only.

Test inventory. The bridge suite contains 86 tests covering: the Clifford relations; $\star^2 = +1$; the symmetry of $C\Gamma^{(5)}$; the gamma-trace rank; the forward rank; the vanishing on the opposite eigenspace; the left inverse; the round trip; the projector property of $\Phi\Phi^-$; equivariance with the character solved for; and the orientation tests of appendix B, including the deliberate branch flip that must break.

D Tensor building blocks

The quadratic block.

$$M_{\mu\nu} \;=\; F_{\mu\alpha\beta\gamma\delta}\, F_\nu{}^{\alpha\beta\gamma\delta}. \tag{D.1}$$

M is symmetric. For self-dual F its trace vanishes, $M^\mu{}_\mu = \langle F, F\rangle = 0$, by the argument of appendix I; this is the same vanishing that fixes the leading degree of the stress-tensor trace, and it means the family $\mathrm{Tr}(M^k)$ starts contributing at $k = 2$.

Quartic blocks. From four copies of F two irreducible structures are needed. We label them $N^{(1050)}$ and $N^{(4125)}$ by the dimensions of the Spin(10) representations in which they transform. They are obtained by projecting the four-fold product onto those irreducible components; the projectors are built from η and ϵ contractions in the fixed index ordering used throughout.

Index placement. All blocks are stored with indices in the down position and raised only at contraction time, so that no expression depends on an implicit metric factor. Contractions are performed in a fixed order determined by the contraction planner, with reduction mod p after every step.

Why the labels matter and where they do not. The representation labels are descriptive. Nothing in the results depends on the irreducible decomposition being the one the labels suggest — the invariants are computed by explicit contraction, and their ranks are what is certified. The labels are retained because they make the candidate inventory of appendix E readable, not because any claim rests on them.

Products versus primitives. At degree ten the product subspace P_{10} is spanned by $I_1^{(4)} I_1^{(6)}$ and $\big(I_1^{(4)}\big)$-type combinations formed from lower-degree invariants; its exact dimension is 2. Everything not in P_{10} is called primitive. The decomposition $A_{10} = G_{10} \oplus P_{10}$ holds with $\dim G_{10} = 12$ and $\dim P_{10} = 2$, summing to 14, and the graph generators are therefore a complement to the products.

E Invariant bases

Identifiers. Candidates carry stable identifiers combining a serial number, a family name and a field degree, as in `c046_portgraph_d12`. The family is one of `portgraph`, `structured` or `trace`. Identifiers are fixed once and never reused, so a candidate referenced in a certificate can always be located. The candidate ordering used in the rank computation is itself hashed and the hash is recorded with every cell, which is what allows cells produced in different working trees to be compared at all.

Degree-four and degree-six. A_4 is one-dimensional, spanned by $\mathrm{Tr}(M^2)$ in the normalisation of appendix D, equivalently by $I_1^{(4)}$. A_6 is two-dimensional; a convenient basis is $\{\mathrm{Tr}(M^3), I_1^{(6)}\}$.

Degree eight. $\dim A_8 = 7$. The tensor-side basis consists of trace monomials in M together with contractions involving the quartic blocks. On the spinor side the corresponding family requires the tensor words discussed in section 7.2: without them the family spans only 6 dimensions.

Degree ten. $\dim_{\mathbb{Q}} A_{10} = 14$, decomposing as

$$A_{10} = G_{10} \oplus P_{10}, \qquad \dim G_{10} = 12, \qquad \dim P_{10} = 2. \tag{E.1}$$

Three subspaces of A_{10} recur:

- B_{10}, the span of the published structures, $\dim_{\mathbb{Q}} B_{10} = 12$;
- D_{10}, the flow-reachable subspace, $\dim_{\mathbb{Q}} D_{10} = 11$;
- P_{10}, the products, $\dim_{\mathbb{Q}} P_{10} = 2$.

Their pairwise incidences are tabulated in table 9.

The Ω and Θ candidates. Two families on the spinor side are written Ω and Θ; they are the tensor words of section 4, distinguished by whether the intermediate tensor structure is quadratic or quartic. They are listed separately because the degree-eight ablation removes exactly these.

The ambiguous published candidates. Two of the published degree-ten expressions cannot be read unambiguously from the source PDF, because bracket colour — which fixes the order of operations — does not survive text extraction. Both readings of each are implemented, denoted by an `-a` or `-b` suffix on the identifier, and both are projected and reported. The compact Q_{10} basis is deliberately chosen to avoid these candidates, so no result in this paper depends on which reading is correct. This is an avoidance, not a resolution; see section 14.

Quotient representatives. The 3 quotient representatives are indexed by the free columns of the rank-11 certificate of appendix H, and are listed in table 6. They are one choice among many: proposition 6 bounds the cardinality of any closing set from below, but says nothing about uniqueness, and we make no canonicality claim for this choice.

F Degree-resolved change-of-basis maps

For each degree at which the two formulations are compared we record an explicit change-of-basis matrix taking the spinor-side basis to the tensor-side basis. The matrices are integer matrices reduced mod p; they are too large to print for degrees eight and ten and are shipped as machine-readable artifacts, but their defining properties are stated here and each is checked.

What is checked for each map. Let T be the tensor-side evaluation matrix on the common samples and S the spinor-side one, at a given degree. The map X satisfies $S = TX$, and the certified properties are:

1. X exists, i.e. the system $S = TX$ is consistent — equivalently $\operatorname{rank} T = \operatorname{rank}[T \mid S]$, which is exactly the statement that the spinor family introduces nothing new;
2. $\operatorname{rank} X$ equals the common rank at that degree;
3. the identity $S = TX$ continues to hold on the 20 held-out samples, which were not used to solve for X.

The third item is the one that carries weight. Solving $S = TX$ on the fitting samples and then reporting that $S = TX$ holds on those same samples would be circular; the held-out check is what makes it a prediction.

Sample families. The 78 common samples divide into 10 sparse, 8 structured, 40 generic and 20 held-out points. The sparse and structured families are included deliberately: a relation that holds only on generic points would be missed by generic sampling alone, and a relation that fails on structured points is a sign that a projector has been mis-specified.

Status by degree.

- Degree four: rank 1, spans equal, holdout passes.
- Degree six: rank 2, spans equal, holdout passes.
- Degree eight: rank 7 for the full family, agreeing at all 4 primes tested. The narrower family without tensor words reaches only 6 and no consistent X exists for it — which is the precise sense in which the words are needed.
- Degree ten: rank 14, spans equal, holdout passes.

G The generic-rank certificate

Coordinate basis. The 126 coordinates are the independent components of a self-dual five-form in a fixed ordering of index five-tuples. All derivatives are taken with respect to these, so the Jacobian is 83×126.

Candidate ordering. The 83 candidates are ordered by a fixed rule and the ordering is hashed. Every cell records the hash, and cells whose hashes differ are not comparable. All 15 cells aggregated here share one ordering hash.

Schedule completeness.

scheduled	83
evaluated	83
errors	0
interrupted	0
silently skipped	0
identically-zero rows	0
Euler homogeneity	*yes*

The distinction between "evaluated" and "scheduled" is tracked explicitly because an earlier stage of this project reported a rank from an incomplete schedule, with the missing candidates invisible in the output.

Prime and sample design. 5 primes $\times$ 3 seeds $=$ 15 cells. The primes are near 2^{15}, chosen so that products of two residues fit in machine integers without overflow; the seeds control the sample point.

Pivots. Every cell returns the same 81 pivot rows and 81 pivot columns, with 0 unstable. The pivot row identities are recorded as candidate identifiers, not as indices, so the comparison is meaningful across runs.

The minor. An explicit 81×81 submatrix is fixed and its determinant computed at prime 32749. The determinant is non-zero, and it is computed by two independent routines — fraction-free Bareiss elimination and a separate exact routine — which agree.

From the minor to characteristic zero. The chosen submatrix has integer entries, being an integer reduction of an analytic derivative evaluated at an integer point. Its determinant is therefore an integer D, and the computation shows $D \not\equiv 0 \bmod p$. Hence $D \neq 0$ over $\mathbb{Z}$, so the corresponding minor over $\mathbb{Q}$ is non-zero and

$$\text{generic functional rank} \;\geq\; 81. \tag{G.1}$$

No prime is excluded and no genericity assumption on the sample point is needed: a specific point exhibiting rank 81 bounds the generic rank from below.

The upper bound. The matching upper bound 81 is analytic, following from the generic orbit dimension of the Lorentz action on Λ^5_+ [20]. We cite it and do not reprove it. The combination of the two gives the exact value, and we are explicit about which half is ours.

Scope. The certified statement is that *the selected family has generic functional rank* 81. It is not the statement that all 83 candidates are algebraically independent: functional independence at a generic point and algebraic independence are different properties, and only the former is established here.

H The exact degree-ten flow closure

Targets and generators. At degrees four through ten the closure consumes exactly the generators

$$\operatorname{Tr}\tau,\ \operatorname{Tr}\tau_2,\ \operatorname{Tr}\tau_3,\ \operatorname{Tr}\tau_4,\ \operatorname{Tr}\tau_5,\ (\operatorname{Tr}\tau)^2,\ \operatorname{Tr}\tau\operatorname{Tr}\tau_2,\ \operatorname{Tr}\tau\operatorname{Tr}\tau_3,\ (\operatorname{Tr}\tau_2)^2,\ \operatorname{Tr}\tau_2\operatorname{Tr}\tau_3. \tag{H.1}$$

This list was determined by instrumenting the closure rather than by reading the construction, so that a generator consumed but not documented would show up.

Activation rules. A target is *generated* when the flow produces it formally, and *activated* when the leading field degree of the generating expression is low enough for it to contribute at the degree under consideration. The leading degrees follow the rule: $\mathrm{Tr}(\tau)$ has leading field degree four — not two, by theorem 7 — while $\mathrm{Tr}(\tau^k)$ has $2k$ for $k \geq 2$, and products add. All generators were checked against this rule independently at every prime, with no exceptions.

Why the distinction is the whole result. The span of *generated* targets has dimension 14, equal to $\dim A_{10}$. The span of *activated* targets, closed under iteration, has dimension 11. Identifying the two returns $\dim Q_{10} = 0$. The raw span is retained in the test suite as a negative fixture for exactly this reason.

Fixed-point algorithm. Starting from the degree-four seed, repeatedly: generate targets from the current span, admit those whose leading degree permits activation, and re-rank. Terminate when a sweep raises no rank. Termination occurred after 3 sweeps.

The exact rational calculation. Of the closure's target rows, 49 were already integral. The remaining 9 were reconstructed as follows: evaluate at 5 fitting primes, combine by the Chinese remainder theorem, and recover a rational by rational reconstruction [26]. The reconstructed rows were then validated at the held-out prime 32771, which took no part in the fit; there were no mismatches. The closure was then run entirely in exact $\mathbb{Q}$ arithmetic.

Dimensions by degree. Over $\mathbb{Q}$ the reachable dimensions are 1, 2, 7 and 11 at degrees four, six, eight and ten.

Lower-bound certificate. An explicit 11×11 integer minor of the reachable span has non-zero determinant. The submatrix, the column set, the integer scale factor by which the rows were cleared of denominators, and the full integer determinant are recorded in the artifact. The determinant was computed by Bareiss elimination [29] and confirmed by an independent exact rational routine.

Upper bound: why none is needed. Running the closure over $\mathbb{Q}$ removes the modular admission test that made the earlier figure a lower bound. At the fixed point, no further generated direction raises the rank over $\mathbb{Q}$; therefore the reached span *is* D_{10}, and $\dim_{\mathbb{Q}} D_{10} = 11$ is an equality. This is the substantive difference between the present computation and its modular predecessor.

Quotient maps. Three exact annihilating covectors are exhibited, each vanishing on D_{10} and independent on A_{10}. They give the quotient map to Q_{10} directly, and the quotient representatives are indexed by the free columns (table 6).

I The free stress-tensor trace

This appendix gives the calculation behind theorem 7 in full, and records how it is verified.

The stress tensor. For a p-form with field strength F of rank p in d dimensions, the free stress tensor takes the form

$$T_{\mu\nu} = \frac{1}{(p-1)!}\, F_{\mu\alpha_2\ldots\alpha_p} F_{\nu}{}^{\alpha_2\ldots\alpha_p} - c\,\eta_{\mu\nu}\,\langle F, F\rangle, \tag{I.1}$$

where c is an improvement coefficient. Different conventions in the literature fix c differently; the point of what follows is that the conclusion does not depend on it.

The trace. Taking the trace of (I.1) with $\eta^{\mu\nu}$, the first term gives $\langle F, F\rangle$ up to normalisation and the second gives $-cd\,\langle F, F\rangle$, so

$$T^{\mu}{}_{\mu} = (1 - cd)\,\langle F, F\rangle. \tag{I.2}$$

Whatever c is, the trace is a multiple of the single scalar $\langle F, F\rangle$. There is no other quadratic Lorentz scalar available: the only way to contract two copies of a five-form into a scalar is the full contraction.

The vanishing. Here $p = 5$ and $d = 10$, so F is a middle-degree form of *odd* degree in an even-dimensional space. Consider the top form $F \wedge F$. Interchanging the two factors of a q-form costs a sign $(-1)^{q^2} = (-1)^q$, which for $q = 5$ is -1; hence $F \wedge F = -F \wedge F$ and $F \wedge F \equiv 0$ identically, for every five-form whatsoever.

Now $\langle F, \star F\rangle\,\mathrm{vol} \propto F \wedge F$, so $\langle F, \star F\rangle = 0$. On either eigenspace of the star, $\star F = \pm F$, giving

$$\langle F, F\rangle = \pm\,\langle F, \star F\rangle = 0. \tag{I.3}$$

By (I.2) the free stress tensor is traceless, for any improvement coefficient. Consequently $\mathrm{Tr}(\tau)$ contributes nothing at field degree two and first contributes at degree four.

Scope of the argument. The derivation uses only self-duality (or anti-self-duality) and the parity of the form degree. It does not use the detailed form of the free dynamics, the choice of improvement term, or the overall normalisation. That robustness is why we regard the input as settled, and why the remaining question for review is narrow: whether the intended free theory and stress convention are the ones fixed in appendix A.

Computational verification. A dedicated test module re-derives the statement from the five-form and the Hodge star alone, importing none of the flow machinery. It checks:

1. $\star^2 = +1$, without which the eigenspace decomposition is not the one assumed;
2. $\langle F, F\rangle = 0$ on self-dual samples, at eight seeds;
3. $\langle F, F\rangle = 0$ on *anti*-self-dual samples, which the argument predicts equally and which would fail if the vanishing were an artefact of the projector rather than of the wedge identity;
4. the control: $\langle F, F\rangle \neq 0$ on an unprojected five-form. This is the test that gives the other three their content, since a quantity that vanished identically would satisfy them all vacuously;

5. the conclusion at four primes;
6. agreement with the production leading-degree table, read once and only to be checked against the independent derivation.

A correction worth recording. An earlier version of the control called the projector routine with an argument believed to select the anti-self-dual space, when in fact that argument selects Lorentzian versus Euclidean signature. The apparent non-vanishing it reported was a Euclidean projector, where $\star^2 = -1$ and the eigenspace decomposition is different. Corrected, both eigenspaces vanish exactly as the argument predicts. The episode is the reason item (1) above is checked explicitly rather than assumed.

The counterfactual. Forcing $\mathrm{Tr}(\tau)$ to begin at degree two activates 24 additional targets and raises the degree-ten reachable dimension from 11 to 14, giving $Q_{10} = 0$ instead of 3. This is retained as a regression fixture: it reproduces precisely the incorrect result this project obtained at an earlier stage, so the test fails loudly if the assumption is ever silently reverted.

J Reconstruction of the published span

The problem. The published degree-ten structures are given as explicit expressions. To compare their span with the product subspace over $\mathbb{Q}$ — rather than modulo a prime — their coordinate vectors must be lifted to $\mathbb{Q}$.

CRT and rational reconstruction. Each published structure was evaluated at 7 fitting primes. The residues were combined by the Chinese remainder theorem into a residue modulo the product, and a rational with bounded numerator and denominator was recovered by rational reconstruction [26]. All 12 published structures lifted; none failed.

Held-out validation. The reconstructed rationals were reduced modulo the held-out prime 32783, which took no part in the fit, and compared against a direct evaluation there. There were no mismatches. This is the step that distinguishes a reconstruction from a guess: rational reconstruction always returns *something*, and only a prime outside the fit can tell you whether it returned the right thing.

Exact intersection. With the lifts in hand,

$$\dim_{\mathbb{Q}} B_{10} = 12, \quad \dim_{\mathbb{Q}} P_{10} = 2, \quad \dim_{\mathbb{Q}}(B_{10} + P_{10}) = 13, \tag{J.1}$$

whence $\dim_{\mathbb{Q}}(B_{10} \cap P_{10}) = 1$ by inclusion–exclusion.

The explicit generator. The one-dimensional intersection is generated by

$$-55\, I^{(1)} - 252\, I^{(2)} + 4200\, I^{(5)} + 5040\, I^{(6)} \;=\; -1040\, I_1^{(4)} I_1^{(6)}, \tag{J.2}$$

an identity between integer combinations, verified at the fresh prime 32869 on 6 independently drawn samples that took no part in either the fit or the holdout.

Normalisation, and a check that failed. The verification initially failed at the fresh prime on all six samples. The cause was in the checking code: the two sides of the identity were each being reduced by their own greatest common divisor before comparison, which destroys precisely the relative scale the identity asserts. Recomputed on a single common scale, the identity holds on all six samples.

We record the failure rather than only the eventual success. A check that fails, is adjusted, and then passes is worthless unless the adjustment is stated, because the adjustment is where the error would hide. The adjustment here was to stop normalising the two sides independently; no coefficient was altered to make the identity fit.

Why the two decompositions differ. The published structures were presented as explicit expressions, not as a basis with specified relations, and there is no reason a set chosen for explicitness should avoid the product sector. That it meets it in exactly one dimension is a structural observation about the published choice, and the direction of the finding — that B_{10} has one dimension *fewer* of primitive content than a naive reading would suggest — can only weaken a claim about the published span, never strengthen one. No result in this paper depends on the equality.

K Algorithms and tests

Graph canonicalisation. Contraction patterns are encoded as vertex-coloured multigraphs and canonicalised with `nauty` [22] through the `pynauty` bindings. Ports are modelled by auxiliary vertices so that the five slots of an F are distinguishable from one another only up to the symmetries the tensor actually has.

An earlier version used a Weisfeiler–Leman style hash as a fallback when the bindings were unavailable. It collided: 49 order-six classes collapsed onto 39 keys, undercounting the atlas. The fallback has been removed rather than fixed. A canonicalisation that is exact at every degree used is a precondition for every dimension in this paper, and a fallback that is silently approximate is worse than no fallback.

Tensor contractions. Contractions are planned once per invariant and then executed with reduction mod p after every step, so intermediates never grow. Planning uses a greedy ordering; `opt_einsum` [24] is supported as an optional accelerator and produces better orderings when present, but it was not installed in the environment that produced the certified results.

Spinor contractions. Spinor-side invariants contract gamma and charge-conjugation matrices along the closed paths of a port graph. The same reduce-after-every-step discipline applies.

Modular arithmetic. Primes are chosen near 2^{15} so that a product of two residues fits in a machine integer with room to accumulate. Reductions are explicit; there is no reliance on a library's implicit modulus. An overflow guard checks that no intermediate exceeds the safe range, and fails rather than wrapping.

What the Euler check is for. Every Jacobian row is tested against Euler's homogeneity relation. It is fair to ask what this catches, since homogeneity of the invariants is guaranteed by the construction. The answer is that the check does not test homogeneity of the invariant; it tests the *differentiation*. Euler's relation ties the computed derivative row back to the value of the invariant at the same point, so a row produced by a mis-amputated contraction — the realistic failure mode when derivatives are taken analytically by removing one copy of F — fails it, whatever rank that row would otherwise have contributed. A row that fails Euler is wrong, and this is the only check in the pipeline that says so about a single row in isolation.

Elimination and determinants. Ranks over $\mathbb{F}_p$ come from exact Gaussian elimination. Determinants certifying a rank are computed twice: by fraction-free Bareiss elimination [29], which keeps every intermediate an integer, and by an independent exact routine. Agreement is required.

Rational reconstruction. CRT combination followed by rational reconstruction [26], always with at least one prime held out of the fit for validation. Sampling arguments elsewhere in the pipeline are Schwartz–Zippel in character [27, 28]: a non-zero polynomial cannot vanish at too many random points, so agreement on independently drawn samples bounds the probability of coincidence, though it does not eliminate it.

Test structure. 254 tensor-side and 86 bridge-side tests. Beyond the ordinary unit tests, the suite contains three categories that carry more weight:

- **Controls.** Tests asserting that a quantity does *not* vanish where it should not, without which the corresponding vanishing tests are vacuous. The $\langle F, F\rangle$ control of appendix I is the clearest example.
- **Counterfactuals.** Tests that deliberately assume something false and assert the consequence — the G-10 counterfactual, the orientation branch flip. These detect an assumption becoming silently unnecessary.
- **Negative fixtures.** Objects retained specifically because they reproduce a historical error, such as the raw target span.

Historical defects, retained as fixtures.

1. Colliding graph canonicalisation, undercounting the atlas.
2. The raw target span identified with the flow closure, giving $Q_{10} = 0$.
3. An unpinned square-root branch in the frame congruence, silently selecting the wrong eigenspace at some primes.
4. A control test calling the projector with a signature flag mistaken for a duality flag.
5. An identity check that normalised the two sides of an equation separately.
6. A rank reported from an incomplete evaluation schedule.

Each is now covered by a test that fails if the defect returns. We list them because the credibility of a computational result rests as much on the errors that were found and fixed as on the checks that passed.

L Reproduction

Environment. See table 12. The dependencies are NumPy [25], `pynauty` (bindings to `nauty` [22]) and `pytest` and `opt_einsum` [24].

`opt_einsum` is *required*, and an earlier version of this appendix described it as optional. It supplies both the contraction order and the intermediate-size and floating-point-operation estimates that the modular contractor enforces as an explicit budget. Without it the code took a naive left-to-right pairwise order with those checks skipped altogether, so the two budgets silently ceased to be budgets. Measured on an 8 GiB host, the degree-eight evaluator then moves from 0.4 s at 0.27 GB peak resident set to an unbounded allocation terminated by the kernel; the failure surfaces as `SIGKILL` with no traceback, which is precisely what such a guard exists to prevent. The fallback now raises rather than running unguarded. Computed values are unaffected: the bridge suite passes identically once it is installed. This was found by a clean-clone reproduction, not known in advance.

Python version. Python 3.10 or newer is required: the source uses PEP 604 union annotations, and on Python 3.9 both test suites fail at collection before any test runs. The certified results were produced under Python 3.13. This requirement was discovered by a clean-clone reproduction rather than known in advance, and it is recorded because on a stock macOS the default `python3` is 3.9.

Obtaining the source.

```
git clone https://github.com/uskutsav-cpu/selfdual-5form-invariants
cd selfdual-5form-invariants
git switch publication/jhep-mentor-draft
python3.13 -m venv .venv
.venv/bin/python3 -m pip install -r requirements.txt
```

Running the test suites.

```
.venv/bin/python3 -m pytest tests -q
.venv/bin/python3 -m pytest spinor_trace_bridge/tests -q
```

Expected: 254 and 86 tests respectively, no failures.

Regenerating the certificates.

```
.venv/bin/python3 scripts/d10_characteristic_zero.py
.venv/bin/python3 scripts/b10_p10_characteristic_zero.py
.venv/bin/python3 scripts/aggregate_rank_matrix.py --self-test
```

The first two are the exact rational computations of appendices H and J. The third re-aggregates the rank matrix; it is read-only with respect to the cells and is order-independent, so it may be run at any time without disturbing them.

Rebuilding the manuscript.

```
.venv/bin/python3 manuscript/scripts/make_numbers.py
.venv/bin/python3 manuscript/jhep/scripts/make_figures.py
.venv/bin/python3 manuscript/jhep/scripts/make_tables.py
cd manuscript/jhep && tectonic -X compile main.tex
```

The macro, figure and table steps must precede the compile; the manuscript reads their output and does not contain any scientific value of its own.

Expected outputs. The headline values are $\dim_{\mathbb{Q}} A_{10} = 14$, $\dim_{\mathbb{Q}} D_{10} = 11$, $\dim_{\mathbb{Q}} Q_{10} = 3$, $\dim_{\mathbb{Q}}(B_{10} \cap P_{10}) = 1$, generic functional rank lower bound 81, bridge rank 126, and 15 rank-matrix cells all of rank 81.

Runtimes and memory. The exact rational closure is the most expensive step, dominated by the rational arithmetic in the final sweeps. Peak resident memory recorded across the rank-matrix cells was under 720 MB. The rank matrix as a whole is the longest job; individual cells are independent and may be run in parallel, but they must not share a row cache — two processes writing one cache is a defect this project has actually experienced.

Determinism, and the limit of it. Figures are byte-identical across repeated runs *within a fixed environment*: matplotlib's PDF creation timestamp is suppressed explicitly, without which otherwise identical figures differ and archive reproducibility fails. Archives normalise member metadata and use a fixed epoch for the same reason.

Byte-identity does *not* survive a change of matplotlib version. A clean-clone run that resolved matplotlib 3.9.4 produced figures visually identical to, and byte-different from, those built with 3.11.1. The plotting library is therefore pinned exactly, in the documentation requirements file, rather than floored. We record this because "byte-reproducible" is otherwise read as a stronger property than it is: what is reproducible is the pipeline given a pinned toolchain, not the bytes under an arbitrary one.

The compiled PDF is the one artifact that is *not* byte-reproducible, and we state this explicitly because the surrounding claims might otherwise be read as covering it. Tectonic writes a fresh trailer `/ID` and build-time metadata on every invocation: three consecutive compilations of identical source in a clean clone produced three distinct SHA-256 digests, differing in about 7,000 bytes once the change propagates into the compressed streams, and setting `SOURCE_DATE_EPOCH` does not suppress it. What is byte-identical across clean rebuilds is everything upstream of the typesetter: all ten figures, the generated tables and macros, and the LaTeX source archive shipped with this draft. A reviewer checking the delivered package should therefore expect the source archive and figure hashes to match exactly and the PDF hash not to; the PDF hash recorded in the build manifest identifies one particular build rather than certifying a reproducible one.

Hashes. The build manifest accompanying this draft records the commit, the SHA-256 of every generated artifact, and the hash of the compiled PDF.

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